PHYSICS

1) A solenoid with an air core has length 48π cm, area of cross section 12 cm^2 and 1200 turns. Self inductance of the solenoid is :-

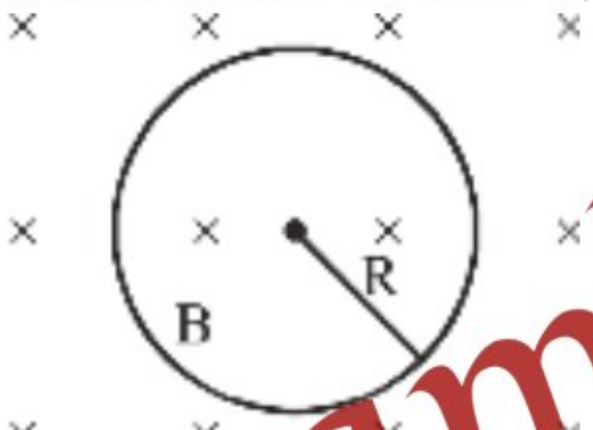
(1) 4.6 mH

(2) 6.9 mH

(3) 1.44 mH

(4) 9.2 mH

2) A conductor loop of radius R is present in a uniform magnetic field B perpendicular the plane of the ring. If radius R varies as a function of time 't' as $R = R_0 + t$. The e.m.f. induced in the loop and



direction of induced current is:-

(1) $2\pi(R_0 + t)B$, clockwise

(2) $\pi(R_0 + t)B$, clockwise

(3) $2\pi(R_0 + t)B$, anticlockwise

(4) Zero

3) A vertical ring of radius r and resistance R falls vertically. It is in contact with two vertical conducting rails which are joined at the top. The rails are without friction and resistance. There is a horizontal uniform, magnetic field of magnitude B perpendicular to the plane of the ring and the



rails. When the speed of the ring is v, the current in the section PQ is

(1) zero

(2) $\frac{2Brv}{R}$

(3) $\frac{4Brv}{R}$

(4) $\frac{8Brv}{R}$

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4) In a transformer ratio of secondary turns (N_S) and primary turn (N_P) i.e. $\frac{N_S}{N_P} = 4$. If the voltage applied on primary side is 200 V, 50 Hz and current is 1 A, then which of the following statements are correct :-

(A) Voltage induced in secondary coil of transformer is 800 V.

(B) If transformer is ideal then current in secondary coil will be 0.25 A.

(C) If efficiency of transformer is 80% then current in secondary coil will be 0.15 A.

(D) Given transformer is step up transformer

(1) A, C, D

(2) A, B, D

(3) A, B, C

(4) A, B, C, D

5) A copper disc of radius 0.1 m rotates about its axis which passes through centre with 10 rev/sec in a uniform magnetic field of 0.1 T which is perpendicular to the plane of disc, then the emf induced between centre and rim of disc is -

(1) $\frac{\pi}{10}$V

(2) 10π mV

(3) $\frac{2\pi}{10}$V

(4) 20π mV

6) A solenoid has turn density 5000 turn/m and cross sectional area 10 cm^2. If a current of 1A flows in it and inside material has relative permeability of 1000, then find energy per unit length of solenoid :-

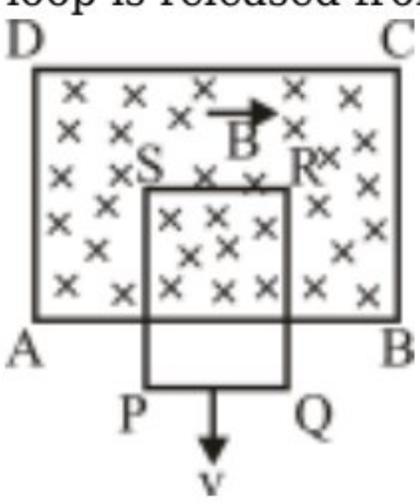
(1) 15.7 J/m

(2) 18.9 J/m

(3) 12.31 J/m

(4) 8.53 J/m

7) A uniform horizontal magnetic field $\vec{B}$ exists in the region ABCD. A rectangular loop of mass m and horizontal side ℓ and resistance R is placed in the magnetic field as shown in the figure. If the loop is released from rest at what velocity it will continue to fall without acceleration?



(1) $\frac{B\ell}{mgR}$

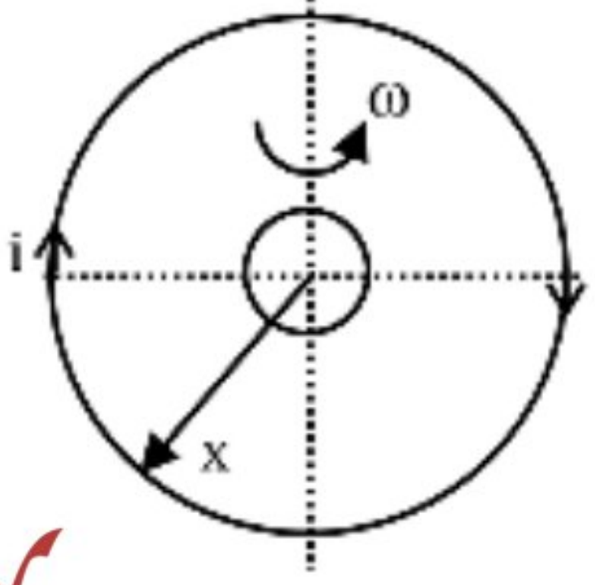
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(2) $\frac{mgR}{B\ell}$

(3) $\frac{B^2\ell^2}{mgR}$

(4) $\frac{mgR}{B^2\ell^2}$

8) A very small circular loop of area A and the resistance R and negligible inductance is initially coplanar and concentric with a much larger fixed loop of radius x. A constant current i is passed in the bigger loop and the smaller loop is rotated with constant angular velocity ω about a diameter



then induced current in the smaller loop as a function of time will be :-

(1) $\frac{\mu_0 iA}{2xR}\sin\ \omega t$

(2) $\frac{\mu_0 iA\omega}{2xR}\sin\ \omega t$

(3) $\frac{\mu_0 iA\omega}{2xR}\sin\ 2\omega t$

(4) Zero

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9) The energy stored in a coil having inductance L and resistance R is U. Now this coil is divided into four equal parts and connected in parallel to same source. The total energy stored in the combination becomes.

(1) $\frac{U}{2}$

(2) $\frac{U}{4}$

(3) $\frac{U}{16}$

(4) 16 U

10) A nonconducting circular ring of radius 4 cm is placed in a time varing magnetic field with rate of 0.2T/s. If 2C charge placed at its circumference then electric force on this charge will be :-

(1) 4×10^{-3} N

(2) 8×10^{-3} N

(3) 6×10^{-2} N

(4) 8×10^{-2} N

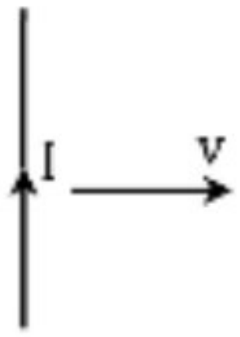
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11) The self-inductance L of a solenoid of length ℓ, area of cross-section A and a fixed number of turns N, increases as

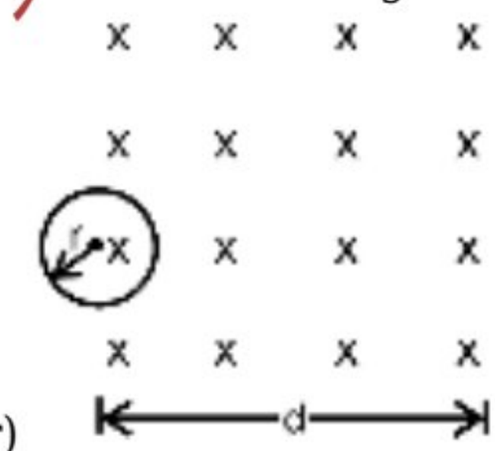
(1) ℓ and A increase.
(2) ℓ decreases and A increases.
(3) ℓ increases and A decreases
(4) both ℓ and A decreases

12) A conducting wire is moving towards right in a magnetic field B. The direction of induced current in the wire is shown in the figure. The direction of magnetic field will be



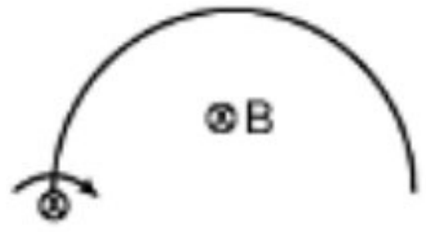
(1) in the plane of paper pointing towards right
(2) in the plane of paper pointing towards left
(3) perpendicular to the plane of paper and inwards
(4) perpendicular to the plane of paper and outwards

13) A conducting loop is pulled with a constant velocity towards a region of uniform magnetic field $\vec{B}$ as shown in the figure. Then, the current involved in the loop is (d > r)



(1) clockwise while entering, anticlockwise while leaving
(2) anticlockwise while entering, clockwise while leaving
(3) clockwise while leaving, clockwise while entering
(4) anticlockwise while leaving, anticlockwise while entering

14) A semicircular wire of radius R is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. There is a uniform magnetic field of strength B perpendicular to plane. The induced e.m.f. between the ends is:



(1) $B\,\omega R^2/2$
(2) $2\,B\,\omega R^2$
(3) is variable
(4) None of these

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15) The magnetic flux linked with a stationary loop of resistance R varies with respect to time during the time period T as follows: $\phi = at(T - t)$. Then amount of heat generated in the loop during that time (inductance of the coil is negligible) is : -

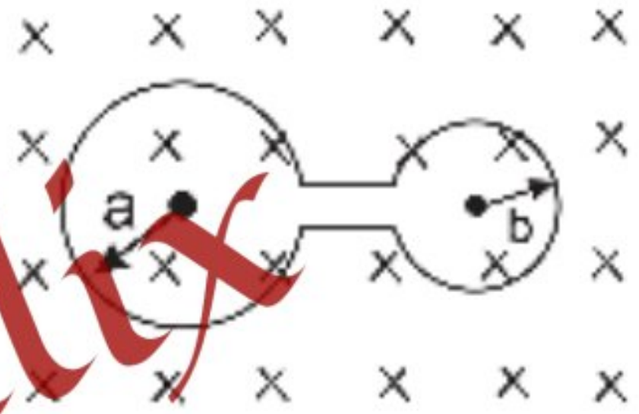
(1) $\frac{aT}{3R}$

(2) $\frac{a^2T^2}{3R}$

(3) $\frac{a^2T^2}{R}$

(4) $\frac{a^2T^3}{3R}$

16) The below figure shows two circular rings of radii a and b (a > b) joined together by wire of negligible resistance. If the arrangement is placed in a time varying magnetic field, $\frac{dB}{dt} = k$ and if the



resistance per unit length of wire is λ, then induced current is

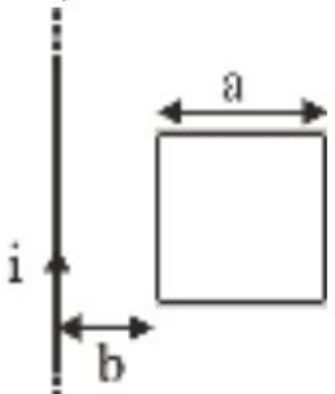
(1) $\frac{k(a+b)}{2\lambda}$

(2) $\frac{k(a-b)}{2\lambda}$

(3) $\frac{k(a^2-b^2)}{2\lambda(a-b)}$

(4) $\frac{k(a^2+b^2)}{2\lambda(a+b)}$

17) Find the mutual inductances between the straight wire and the square loop shown in figure.



(1) $\frac{\mu_0 a}{2\pi}\ell n\left(1+\frac{a}{b}\right)$

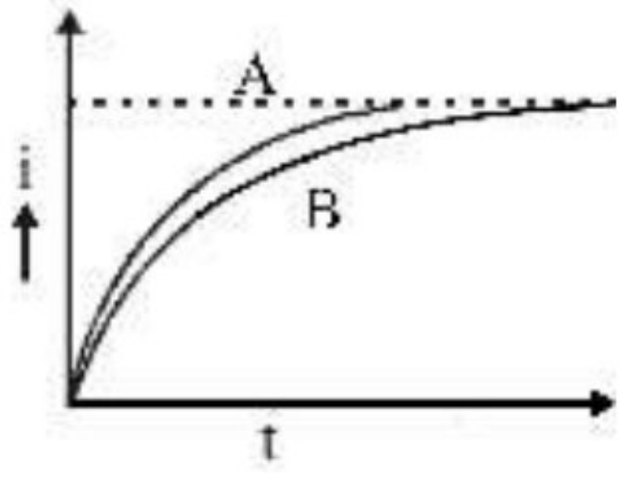
(2) $\frac{\mu_0 a}{2\pi}\ell n\left(1-\frac{a}{b}\right)$

(3) $\frac{\mu_0 a}{4\pi}\ell n\left(1+\frac{a}{b}\right)$

(4) $\frac{\mu_0 a}{8\pi}\ell n\left(1+\frac{a}{b}\right)$

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18) A circuit consisting of a constant emf E, a self-inductance L and a resistance R is closed at time t = 0. The relation between the current i in the circuit and the time t is as shown by the curve 'A' in the figure. When one or more of parameters E, R and L are changed, the curve 'B' is obtained. Then



it is possible that :-

(1) E and R are kept constant and L is increased

(2) E and R are kept constant and L is decreased

(3) E and L are kept constant and R is decreased

(4) E and L are kept constant and R is increased

19) Two small circular coils are made of copper wires of same length. In the first coil the number of turns is 3n and radius is r. If in the second coil number of turns is n and radius is 3r, the ratio of self inductances of the coils will be :–

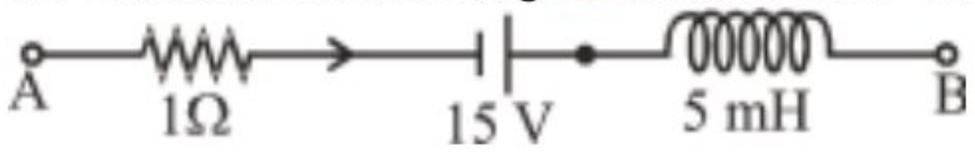
(1) 9 : 1

(2) 3 : 1

(3) 1 : 3

(4) 1 : 9

20) The network shown in the figure is a part of a complete circuit. If at a certain instant the current i is 5 A and is decreasing at the rate of 10^3 A/s then $V_B - V_A$ is :-



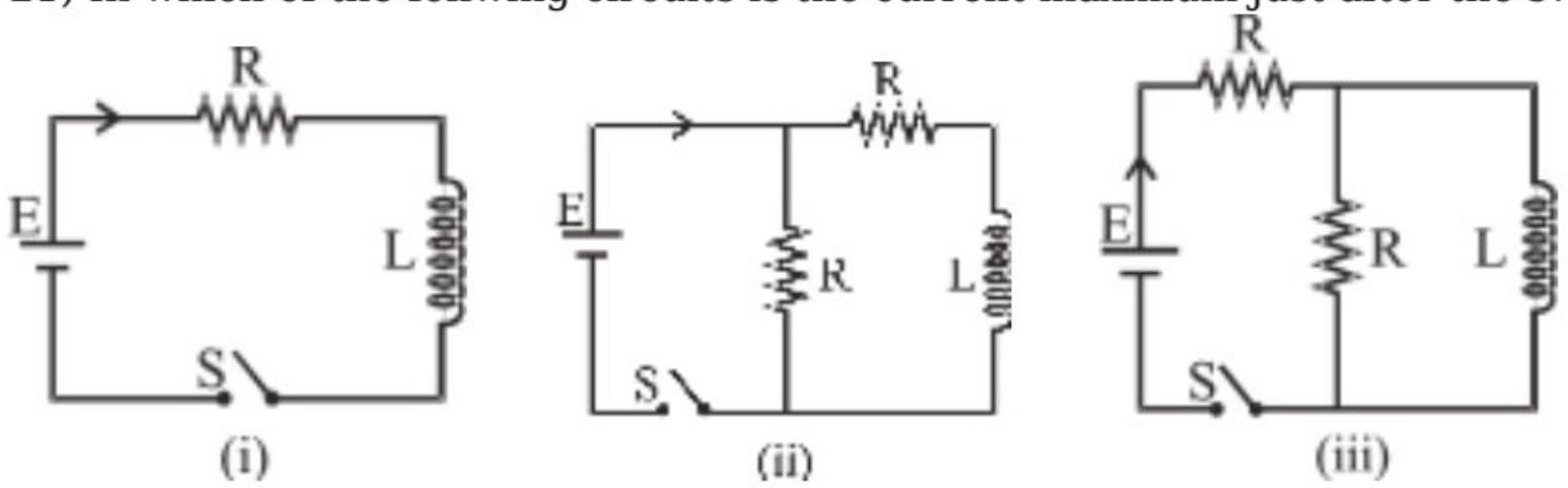
(1) 5 V

(2) 10 V

(3) 15 V

(4) 20 V

21) In which of the follwing circuits is the current maximum just after the switch S is closed :-



(1) (i)

(2) (ii)

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(3) (iii)

(4) Both (ii) and (iii)

22) The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m^2 acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1μV, is :-

(1) $\frac{2}{\pi}$cm

(2) $\frac{3}{\pi}$cm

(3) $\frac{4}{\pi}$cm

(4) $\frac{5}{\pi}$cm

23) The mutual inductance between a primary and secondary circuits is 0.5 H. The resistance of the primary and the secondary circuits are 20Ω and 5Ω respectively. To generate a current of 0.4 A in the secondary, current in the primary must be changed at the rate of :-

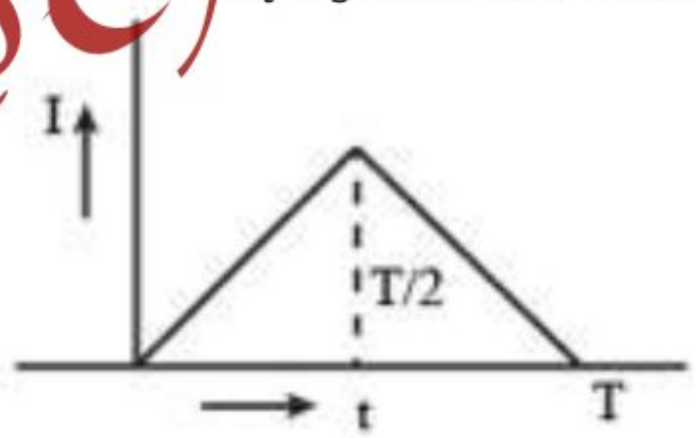
(1) 4.0 A/s

(2) 1.6 A/s

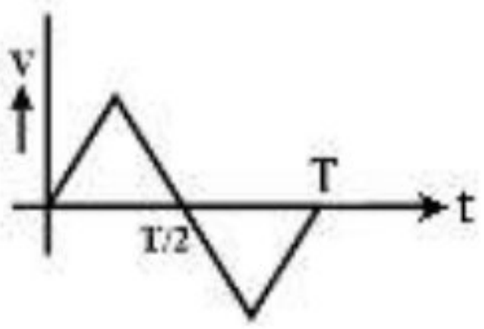
(3) 16.0 A/s

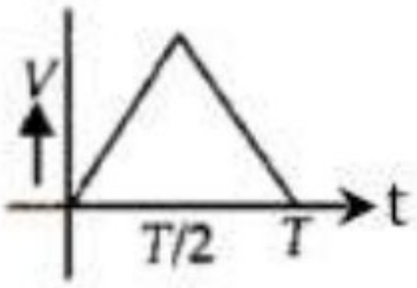
(4) 8.0 A/s

24) The current (I) in the inductance is varying with time according to the plot shown in the figure.



Which one of the following is the correct variation of voltage with time in the coil?

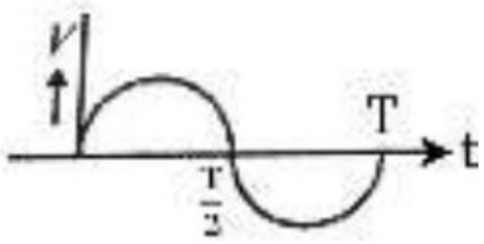
(1)


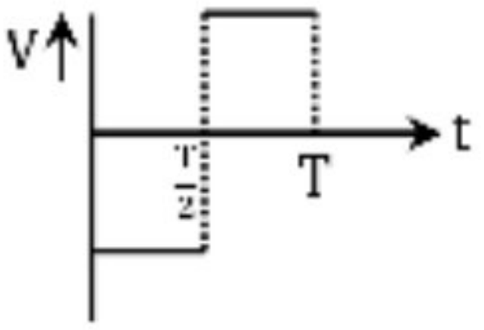
(2)


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(3) 

(4) 

25) **Statement I :** Eddy currents are used in electromagnetic Induction brakes.
Statement II : Eddy currents always opposes the relative motion.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

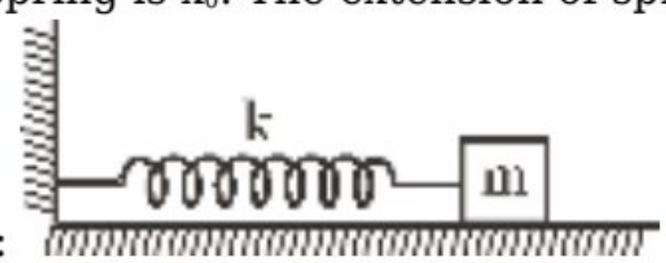
26) A plank P and block Q are arranged as shown on a smooth table top. They are given velocities 3 m/s and 6 m/s respectively. The length of plank is 1m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. The final velocity of plank then is (coefficient of friction between plank and block is 0.3 and mass of plank is double of block) :- 

(1) 4 m/s

(2) 5 m/s

(3) 4.5 m/s

(4) zero

27) A block of mass m is connected to a light spring of spring constant k. The block is released when the extension of the spring is x_0. The extension of spring when the power due to spring force on the block is maximum is : 

(1) $\frac{x_0}{2}$

(2) $\frac{x_0}{\sqrt{2}}$

(3) $\frac{x_0}{\sqrt{3}}$

(4) x_0

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28) A boat of length 10m and mass 450 kg is floating without motion in still water. A man of mass 50 kg standing at one end of it, walks to the other end of it and stops. The magnitude of the displacement of the boat in meter relatively to ground is :-

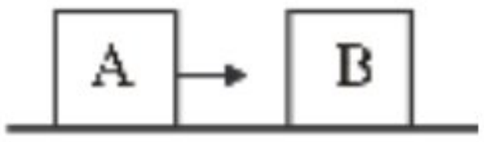
(1) Zero

(2) 1 m

(3) 2m

(4) 5m

29) Two identical blocks A and B are placed on a fixed horizontal surface which is perfectly smooth. The block B is at rest while the block A is approaching towards B with a uniform velocity of 10 m/s. The coefficient of restitution for all the collisions is 0.5. The block B just after the collision moves with the speed of :



(1) 3.75 m/s

(2) 5.625 m/s

(3) 7.5 m/s

(4) zero

30) Two particles having position vectors $\vec{r}_1 = (3\hat{i} + 5\hat{j})m$ and $\vec{r}_2 = (-5\hat{i} - 3\hat{j})m$ are moving with velocities $\vec{v}_1 = 4\hat{i} + 3\hat{j}$ and $\vec{v}_2 = (a\hat{i} + 7\hat{j})$ m/s. If they collide after 2 seconds, the value of a is :-

(1) 2

(2) 4

(3) 6

(4) 8

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31) A bullet of mass 100 g is fired from below into the bob of mass 400 g of a long simple pendulum as shown in figure. The bullet remains inside the bob & the bob rises through a height of 5m. The speed of bullet is (Take g = 10 m/s^2):



(1) 60 m/s

(2) 45 m/s

(3) 50 m/s

(4) 55 m/s

32) Ball 1 and 2 of masses 4 kg & 2 kg respectively are moving towards each other with velocities 10 m/s & 5 m/s undergo a head on collision. After the collision, the velocity of ball 2 is observed to be 7 m/s to the right then coefficient of restitution is:

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(1) $e = \frac{1}{2}$

(2) $e = \frac{1}{4}$

(3) $e = \frac{1}{3}$

(4) $e = \frac{1}{5}$

33) A bomb of mass 9kg initially at rest explodes into two piece of masses 3 kg and 6 kg. The kinetic energy of the 3 kg mass is 216 J. The kinetic energy of the 6kg mass will be :-

(1) 213 J

(2) 108 J

(3) 432 J

(4) 54 J

34) A 1 kg stationary bomb is exploded in three parts having mass ratio 4 : 4 : 7. Parts having same mass move in perpendicular directions with velocity 70 m/s, then the velocity of bigger part will be :-

(1) $40\sqrt{2}$ m/s

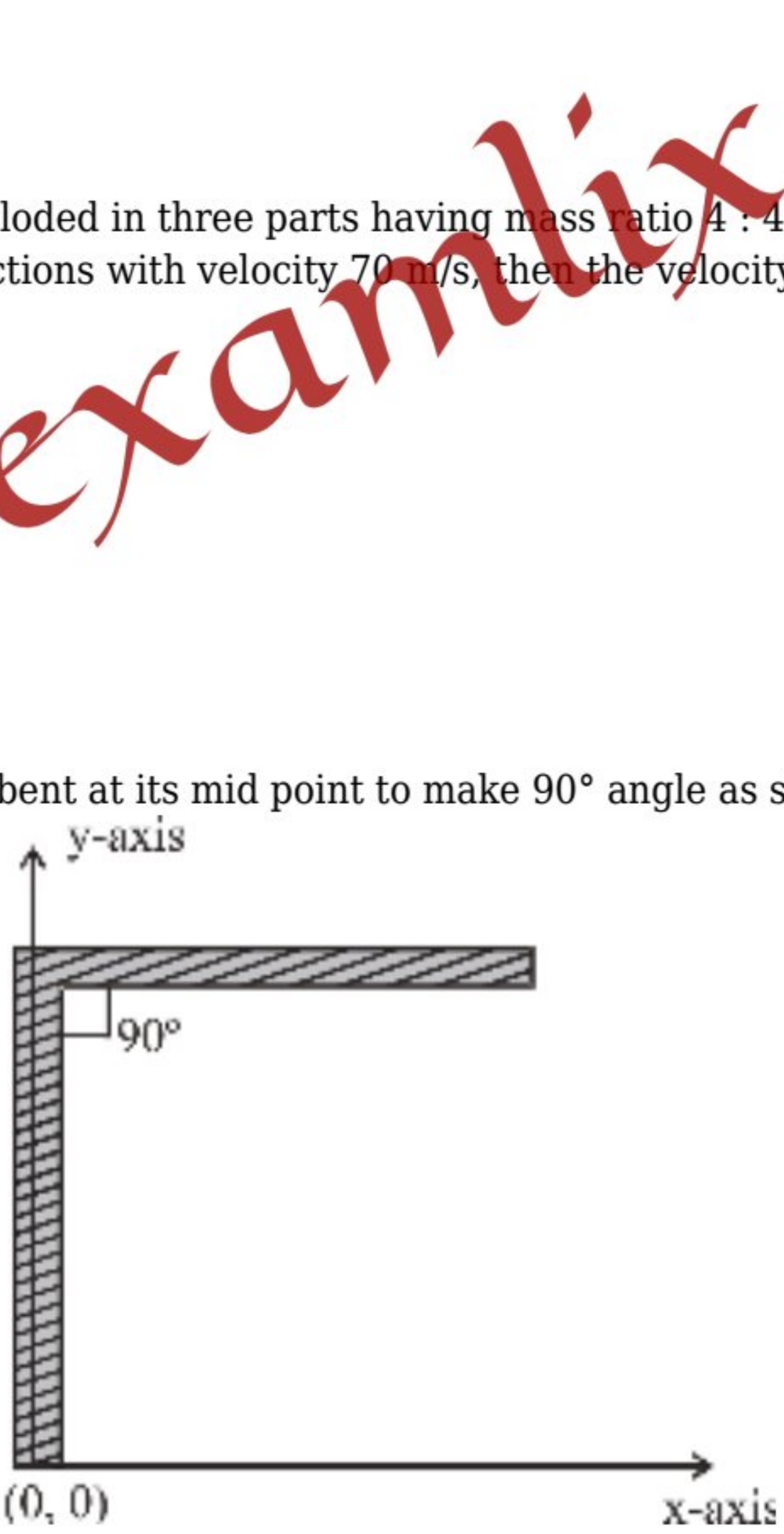
(2) $\frac{40}{\sqrt{2}}$ m/s

(3) $20\sqrt{2}$ m/s

(4) $\frac{20}{\sqrt{2}}$ m/s

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35) A uniform rod of length 2□ is bent at its mid point to make 90° angle as shown in figure. Co-



ordinates of centre of mass are :-

(1) $\left(\frac{2\ell}{3}, \frac{\ell}{3}\right)$

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(2) $\left(\frac{\ell}{8}, \frac{\ell}{8}\right)$

(3) $\left(\frac{\ell}{4}, \frac{3\ell}{4}\right)$

(4) $\left(\frac{\ell}{3}, \frac{\ell}{6}\right)$

36) A body A, of mass m = 0.1 kg has an initial velocity of $3\hat{i}$ m/s. It collides elastically with another body B of the same mass which has an initial velocity of $5\hat{j}$ m/s. After collision, A moves with a velocity $\vec{V} = 4(\hat{i} + \hat{j})$. The energy of B after collision is written as $\frac{x}{10}$J. The value of x is –

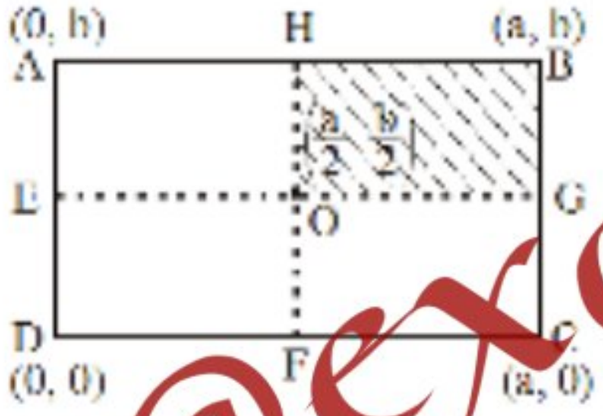
(1) 1

(2) 2

(3) 3

(4) 4

37) A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be :-



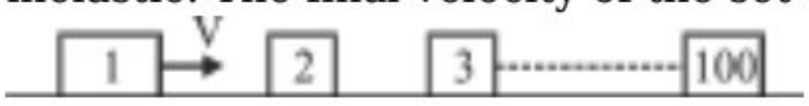
(1) $\left(\frac{2a}{3}, \frac{2b}{3}\right)$

(2) $\left(\frac{5a}{3}, \frac{5b}{3}\right)$

(3) $\left(\frac{3a}{4}, \frac{3b}{4}\right)$

(4) $\left(\frac{5a}{12}, \frac{5b}{12}\right)$

38) There are hundred identical blocks equally spaced on a frictionless track as shown. Initially all the blocks are at rest blocks 1 is pushed with velocity V towards blocks 2. All collisions are perfectly inelastic. The final velocity of the set of hundred stucked blocks will be :-



(1) $\frac{V}{99}$

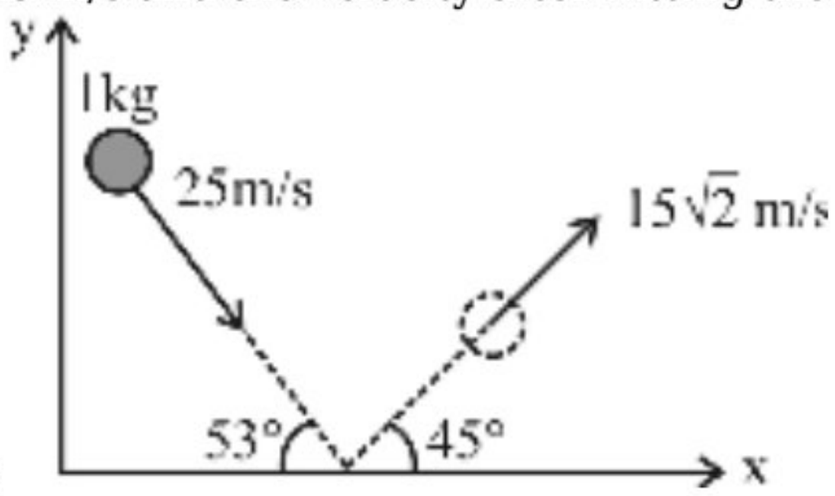
(2) $\frac{V}{100}$

(3) V

(4) Zero

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39) A particle of mass 1 kg bounces against the ground as shown in the figure. The approaching velocity is 25 m/s and the velocity after hitting the ground is $15\sqrt{2}$m/s, the impulse exerted on the particle is :



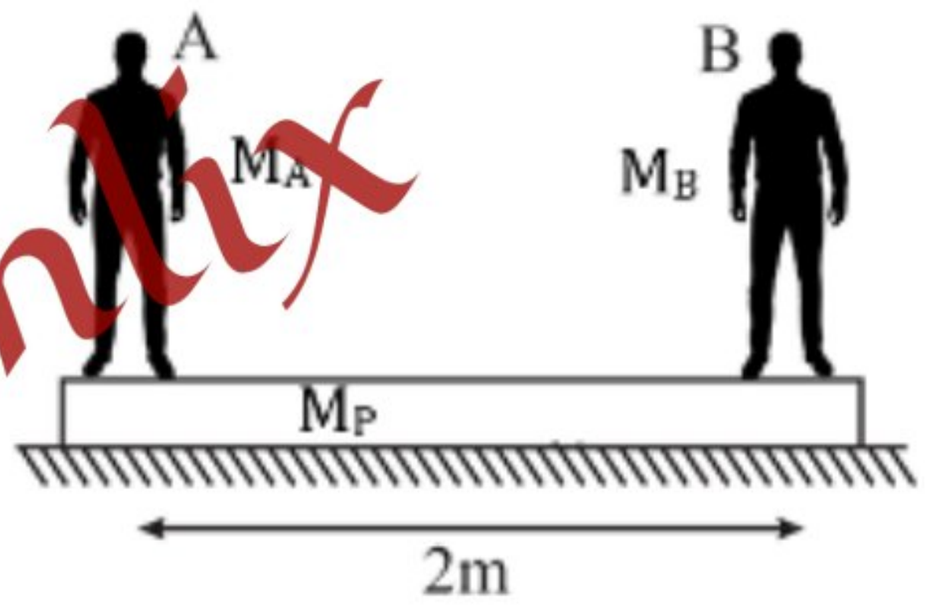
(1) 5 Ns

(2) $5\left(5-3\sqrt{2}\right)$ Ns

(3) 35 Ns

(4) None of these

40) In the figure shown, when the person A and B exchange their positions, then match the List–I and II. [There is no friction between plank and ground]



List–I		**List–II**	
(P)	The distance (in m) moved by the center of mass of the system is	(1)	0.2
(Q)	The distance (in m) moved by the plank is	(2)	1.8
(R)	The distance (in m) moved by A with respect to ground is	(3)	0
(S)	The distance (in m) moved by B with respect to ground is	(4)	2.2
		(5)	1.2
		(6)	2.8

M_A = 90 kg, M_B = 120 kg, M_P = 90 kg

(1) (P)-3, (Q)-1, (R)-4; (S)-2

(2) (P)-5, (Q)-4, (R)-3; (S)-2

(3) (P)-6, (Q)-1, (R)-2; (S)-4

(4) (P)-2, (Q)-1, (R)-6; (S)-5

41) A sphere of mass m_1 = 3 kg collides with a velocity of 7 m/s directly on another sphere of mass m_2 = 5kg. Their velocities after the impact are in the ratio 2 : 3. If the coefficient of restitution is e, find $\frac{1}{e}$.

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(1) 3

(2) 5

(3) 7

(4) 9

42) A thin rod of length 6 m is lying along the x-axis with its ends at x=0 and x = 6 m. Its linear density (mass/length) varies with x as kx^4. Find the position of centre of mass of rod in meters.

(1) 5 m

(2) 3 m

(3) 2 m

(4) 4 m

43) Particles of masses m, 2m, 3m.....nm grams are placed on the same line at distances ℓ, 2ℓ, 3ℓ,.....nℓ cm from a fixed point. The distance of centre of mass of the particles from the fixed point in centimeters is:

(1) $\frac{(2n+1)\ell}{3}$

(2) $\frac{\ell}{n+1}$

(3) $\frac{n(n^2+1)\ell}{2}$

(4) $\frac{2\ell}{n(n^2+1)}$

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44) A shell following a parabolic path explodes some where in its flight. The centre of mass of fragments will move in :-

(1) Vertical direction

(2) Any direction

(3) Horizontal direction

(4) Same parabolic path

45) Which of the following statements is false?

(1) The magnitude of momentum of a heavy object is greater than that of a light object moving at the same speed

(2) In a perfectly inelastic collision, all the initial kinetic energy of the colliding bodies is dissipated

(3) The momentum of a system of colliding bodies may be conserved even though the total mechanical energy may not

(4) The velocity of the center of mass of a system is the system's net momentum divided by its total mass

CHEMISTRY

1) The occurrence of a reaction is impossible if :-

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(1) ΔH is positive; ΔS is negative

(2) ΔH is negative; ΔS is also negative but $\Delta H > T\Delta S$

(3) ΔH is negative; ΔS is positive

(4) ΔH is positive; ΔS is also positive but $\Delta H < T\Delta S$

2) Internal energy change during a reversible isothermal expansion of an ideal gas is :-

(1) Always negative

(2) Always positive

(3) Zero

(4) May be positive or negative

3) The difference in ΔH and ΔE for the combustion of methane at 25°C would be :–

(1) $-2 \times 298 \times 2$ Cals.

(2) Zero

(3) $2 \times 25 \times -3$ Cals.

(4) $2 \times 298 \times -3$ Cals.

4) For the gaseous reaction involving the complete combustion of isobutane -

(1) $\Delta H = \Delta E = 0$

(2) $\Delta H < \Delta E$

(3) $\Delta H = \Delta E$

(4) $\Delta H > \Delta E$

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5) **Assertion :-** There is no change in internal energy of an ideal gas at constant temperature.

Reason :- Internal energy of an ideal gas is exclusively function of temperature only.

(1) Assertion is True but the Reason is False.

(2) Both assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(4) Both Assertion & Reason are False.

6) The heat of vaporisation of water at 100°C is 540 cal g^{-1}. Calculate the entropy change when one mole of water at 100°C is vaporised :-

(1) 367 cal K^{-1} mol^{-1}

(2) 26.05 cal K^{-1} mol^{-1}

(3) 5.40 cal K^{-1} mol^{-1}

(4) 1.44 cal K^{-1} mol^{-1}

7) 5 mole of an ideal gas expand reversibly from a volume of 8 dm^3 to 80 dm^3 at a temperature of 27°C. The change in entropy is :–

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(1) – 41.57 JK^{-1}

(2) 95.73 JK^{-1}

(3) – 95.73 JK^{-1}

(4) 41.57 JK^{-1}

8) The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :

(1) 48 J

(2) –32 J

(3) 32 J

(4) 40 J

9) For the reaction $Ag_2O(s) \rightarrow 2Ag(s) + \frac{1}{2} O_2(g)$, which one of the following is true :

(1) $\Delta H = \Delta E$

(2) $\Delta H = \frac{1}{2} \Delta E$

(3) $\Delta H > \Delta E$

(4) $\Delta H < \Delta E$

10) Two moles of an ideal gas expand spontaneouly into vacuum. The work done is :–

(1) 2 J

(2) Zero

(3) 8 J

(4) 4 J

11) For which reaction from the following, ΔS will be maximum ?

(1) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

(2) $C(s) + O_2(g) \rightarrow CO_2(g)$

(3) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(4) $Ca(s) + \frac{1}{2} O_2(g) \rightarrow CaO(s)$

12) The difference between heat of reaction at constant pressure (ΔH) and at constant volume (ΔE) for the reaction :

$C_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{(g)}$ is :-

(Assume R = 0.002 kcal K^{-1} mol^{-1} and temperature = T K)

(1) $-\frac{T}{2}$cal

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(2) $\frac{T}{2}$cal

(3) T cal

(4) –T cal

13) One mole of a non-ideal gas undergoes a change of state from (1 atm, 3L, 200 K) to (4 atm, 5L, 250 K) with a change in internal energy (ΔU) = 40 L-atm. The change in enthalpy of the process in L-atm :-

(1) 57

(2) 43

(3) 42

(4) None

14) Calculate the work done when 10 mol of ideal gas is compressed reversibly and isothermally from 10 L to 1 L at 27°C ?

(1) –1013 J

(2) –57.44 kJ

(3) 1013 J

(4) 57.44 kJ

15) For a sample of 'n' moles of perfect gas when its pressure is changed isothermally from p_i to p_f, the entropy change is given by

(1) $\Delta S = RT \ln \left(\frac{p_i}{p_f}\right)$

(2) $\Delta S = nRT \ln \left(\frac{p_f}{p_i}\right)$

(3) $\Delta S = nR \ln \left(\frac{p_i}{p_f}\right)$

(4) $\Delta S = nR \ln \left(\frac{p_f}{p_i}\right)$

16) The work done in ergs for a reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25°C is :

(1) $-2.303 \times 0.0821 \times 298 \log 0.5$

(2) $-2.303 \times 2 \times 298 \log 2$

(3) $-2.303 \times 8.31 \times 10^7 \times 298 \log 2$

(4) $-2.303 \times 0.0821 \times 298 \log 2$

17) The enthalpy of vaporisation of liquid water using the data

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$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; $\Delta H° = -285.77$ kJ/mol

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta H° = -241.84$ kJ/mol

is

(1) 527.61 kJ/mol

(2) –43.93 kJ/mol

(3) –527.61kJ/mol

(4) +43.93 kJ/mol

18) What is $\Delta G°$ for this reaction?

$\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightleftharpoons NH_3(g)$; $K_P = 4.42 \times 10^4$ at 25° C

(1) –11.5 kJ mol^{-1}

(2) –26.5 kJ mol^{-1}

(3) –2.2 kJ mol^{-1}

(4) –0.97 kJ mol^{-1}

19) If enthalpies of formation for $C_2H_4(g)$, $CO_2(g)$ and $H_2O(l)$ at 25°C and 1 atm pressure be 52, –394 and –286 kJ mol^{-1} respectively, enthalpy of combustion of $C_2H_4(g)$ will be:

(1) –141.2 kJ mol^{-1}

(2) –1412 kJ mol^{-1}

(3) 141.2 kJ mol^{-1}

(4) +1412 kJ mol^{-1}

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20) 5 moles of an ideal gas at 27 °C expands isothermally and reversibly from a volume of 6 L to 60 L. The work done in kJ is

(1) –14.7

(2) –56.72

(3) +28.72

(4) –28.72

21) Following reaction is occurring in an automobile

$2C_8H_{18}(g) + 25O_2(g) \rightarrow 16CO_2(g) + l8H_2O(g)$

The sign of ΔH, ΔS and ΔG would be :

(1) –, +, –

(2) +, –, +

(3) +, +, –

(4) –, +, +

22) Standard Gibbs free energy change $\Delta G°$ for a reaction is zero. The value of the equilibrium constant will be:

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(1) ∞

(2) 100

(3) 1

(4) 10

23) The relation of ΔH and ΔU is represented as :

(1) $\Delta H = \Delta U - \Delta nRT$

(2) $\Delta U = \Delta V + \Delta H$

(3) $\Delta H = \Delta U - P\Delta V$

(4) $\Delta U = \Delta H - P\Delta V$

24) The correct relation between standard Gibb's energy change (ΔG°) and Equilibrium constant (K_p) is :

(1) $\Delta G° = RT \log_e K_P$

(2) $\Delta G° = RT^2 \square nK_P$

(3) $-\Delta G° = RT \log_e K_P$

(4) None

25) For a liquid, enthalpy of fusion is 1.435 kcal mol^{-1} and molar entropy change is 5.26 cal mol^{-1} K^{-1}. The melting point of the liquid is:

(1) 173 K

(2) 0° C

(3) –273° C

(4) 100° C

26) For spontaneous process:

(1) $\Delta S_{total} < 0$

(2) $\Delta S_{total} > 0$

(3) $\Delta S_{total} = 0$

(4) none of these

27) At 0°C, ice and water are in equilibrium and $\Delta H = 6$ kJ mol^{-1}. For this process : $H_2O(s) \rightleftharpoons H_2O(l)$
The values of ΔS and ΔG for conversion of ice to liquid water at 0°C are

(1) 0.219 J K^{-1} mol^{-1} and 0

(2) 2.19 J K^{-1} mol^{-1} and 0

(3) 0.0219 J K^{-1} mol^{-1} and 0

(4) 21.9 J K^{-1} mol^{-1} and 0

28) Standard enthalpy and standard entropy changes for the oxidation of ammonia at 298 K are –382.64 kJ mol^{-1} and –145.6 JK^{-1} mol^{-1}, respectively. Standard Gibbs energy change for the same

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reaction at 298 K is :

(1) –339.3 kJ mol^{-1}

(2) –439.3 kJ mol^{-1}

(3) –523.2 kJ mol^{-1}

(4) –221.1 kJ mol^{-1}

29) Given that:

$2Fe(s) + \frac{3}{2}O_2(g) \rightarrow Fe_2O_3(s)$ ($\Delta H = -193.4$ kJ) ...(i)

$Mg(s) + \frac{1}{2}O_2(g) \rightarrow MgO(s)$ ($\Delta H = -140.2$ kJ) ...(ii)

What is ΔH of the reaction?

$3Mg + Fe_2O_3 \rightarrow 3MgO + 2Fe$

(1) 272.3 kJ

(2) 227.2 kJ

(3) –272.3 kJ

(4) –227.2 kJ

30) Determine the heat of the following reaction :

$FeO(s) + Fe_2O_3(s) \rightarrow Fe_3O_4(s)$

Given information :

$2Fe(s) + O_2(g) \rightarrow 2FeO(s)$; $\Delta H° = -544$ kJ

$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$; $\Delta H° = -1648.4$ kJ

$Fe_3O_4(s) \rightarrow 3Fe(s) + 2O_2(g)$; $\Delta H° = 1118.4$ kJ

(1) –22.2 kJ

(2) –1074 kJ

(3) +249.8 kJ

(4) +2214.6 kJ

31) For the reaction,

$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$; $\Delta H = -571$ kJ

bond energy of (H–H) = 435 kJ; of (O=O) = 498 kJ;

then calculate the average bond energy of (O–H) bond using the above data.

(1) 271 kJ

(2) –484 kJ

(3) 484 kJ

(4) –271 kJ

32) Use the bond energies in the table to estimate ΔH for this reaction:

$CH_2=CH_2 + Cl_2 \rightarrow ClCH_2–CH_2Cl$

Bond	C – C	C = C	C – Cl	C – H	Cl – Cl
Bond energy (kJ/mol)	347	612	331	414	243

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(1) ΔH = –684 kJ

(2) ΔH =+177 kJ

(3) ΔH = –154 kJ

(4) ΔH = +89 kJ

33) Oxidation states of chlorine in $HClO_4$ and $HClO_3$ are : -

(1) +3, +4

(2) +7, +5

(3) +5, +7

(4) +4, +3

34) If an element is in its lowest oxidation state, under proper conditions, it can act as : -

(1) a reducing agent

(2) an oxidising agent

(3) oxidising as well as reducing agent

(4) neither oxidising nor reducing agent

35) In which one of the following changes, there is transfer of five electrons?

(1) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

(2) $CrO_4^{2-} \rightarrow Cr^{3+}$

(3) $MnO_4^- \rightarrow MnO_2$

(4) $MnO_4^- \rightarrow Mn^{2+}$

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36) In the following reaction,
$4P + 3KOH + 3H_2O \rightarrow 3KH_2PO_2 + PH_3$

(1) P is oxidised only

(2) P is oxidised as well as reduced

(3) P is reduced only

(4) none of the above

37) The oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$ is:

(1) +1

(2) +2

(3) +3

(4) 0

38) For the redox reaction,
$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$
the correct coefficients of the reactants for the balanced equation are:

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NOVA

	MnO_4^-	$C_2O_4^{2-}$	H^+
(1)	5	16	2
(2)	2	5	16
(3)	2	16	5
(4)	16	5	2

(1) 1

(2) 2

(3) 3

(4) 4

39) The value of n in the following equation is:

$Cr_2O_7^{2-} + 14H^+ + nFe^{2+} \rightarrow 2Cr^{3+} + nFe^{3+} + 7H_2O$

(1) 7

(2) 6

(3) 2

(4) 3

40) Which of the following shows nitrogen with its increasing order of oxidation number ?

(1) $NO < N_2O < NO_2 < NO_3^- < NH_4^+$

(2) $NH_4^+ < N_2O < NO < NO_2 < NO_3^-$

(3) $NH_4^+ < N_2O < NO_2 < NO_3^- < NO$

(4) $NH_4^+ < NO < N_2O < NO_2 < NO_3^-$

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41) What are the stoichiometric coefficient of the following reaction ?

$pI_2 + qOH^- \rightarrow IO_3^- + rI^- + 3H_2O$

(1) $p \rightarrow 6, q \rightarrow 3, r \rightarrow 5$

(2) $p \rightarrow 3, q \rightarrow 2, r \rightarrow 3$

(3) $p \rightarrow 3, q \rightarrow 3, r \rightarrow 3$

(4) $p \rightarrow 3, q \rightarrow 6, r \rightarrow 5$

42) Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : In the dissociation of PCl_5 at constant pressure and temperature addition of helium at equilibrium increases the dissociation of PCl_5.

Reason (R) : Equilibrium constant changes only with change in temperature.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

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(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

43)

Correct match list X and Y

	List X		**List Y**
(A)	Active mass	(I)	$\Delta n_g = 0$
(B)	Dynamic nature	(II)	Molar concentration
(C)	A + heat $\rightleftharpoons$ B	(III)	Vant hoff's equation
(D)	$\log (K_2/K_1) = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$	(IV)	Shifting forward direction if temperature increases
(E)	$2A_{(g)} + B_{(g)} \rightleftharpoons 3C_{(g)}$	(V)	Chemical equilibrium

(1) A - (V), B - (II), C - (III), D - (I), E - (IV)

(2) A - (V), B - (IV), C - (III), D - (II), E - (I)

(3) A - (II), B - (V), C - (IV), D - (III), E - (I)

(4) A - (II), B - (V), C - (III), D - (I), E - (IV)

44) Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): ΔH_f° for $H_2(g)$, and $Cl_2(g)$ is zero.

Reason (R): ΔH_f° for every element in its stable and standard state is zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

45) Given below are two statements:

Statement I : In an isolated system there is no exchange of energy but exchange of matter is possible between system and the surroundings.

Statement II : The presence of reactants in a thermous flask is an example of an isolated system.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct.

(4) Both statement I and statement II are correct.

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BIOLOGY

1) During coagulation of blood, fibrins are formed by the conversion of inactive fibrinogen in the plasma by the enzyme–

(1) Prothrombin
(2) Thrombokinase
(3) Thrombin
(4) Carbonic anhydrase

2) Origin of heart beat and its conduction is represented by

(1) AV node → Bundle of His → SA node → Purkinje fibres
(2) SA node → Purkinje fibres → AV node → Bundle of His
(3) Purkinje fibres → SV node → AV node → Bundle of His
(4) SA node → AV node → Bundle of His → Purkinje fibres

3) The pace maker in frog heart is :-

(1) Sinus venosus
(2) AV Node
(3) Conus arteriosus
(4) Heart muscles

4) AB blood group can be donated to :–

(1) A
(2) B
(3) AB
(4) O

5) Which among the following is correct during each cardiac cycle ?

(1) The volume of blood pumped out by the right and left ventricles is same.
(2) The volume of blood pumped out by the right and left ventricles is different
(3) The volume of blood received by each atrium is different
(4) The volume of blood received by the aorta and pulmonary artery is different

6) **Assertion :** High blood pressure leads to heart disease and also affects vital organs like brain and kidney.
Reason : On Repeated checks of BP of an individual is 140/90 or more represents hypertension.

(1) (A) and (R) both are true and (R) is correct explanation of (A).
(2) (A) and (R) both are correct but (R) is not correct explanation of (A)
(3) (A) is true (R) is false

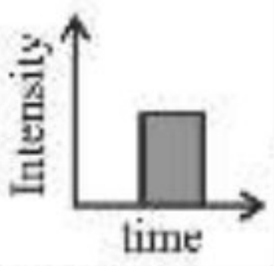
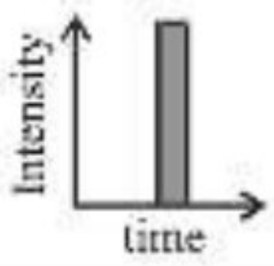
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(4) (A) and (R) both are incorrect.

7) Which animal has most mixing of oxygenated and deoxygenated blood in the ventricles ?

(1) Scoliodon
(2) Rabbit
(3) Frog
(4) Human

8) Choose the correct statements from the differences given below :-

	I Heart Sound		II Heart Sound
A.	LUBB 	A.	DUB 
B.	due to closure of semilunar valves	B.	due to closure of AV valves
C.	Diastolic sound	C.	Ventricular (Systolic) Sound

(1) 'A'
(2) A and B
(3) B and C
(4) A, B and C

9) How many organs or structure in human beings receive oxygenated as well as deoxygenated blood ?
Lungs, Heart, Kidney, Liver, Pancreas, Gonads, Spleen, Ear, Hypothalamus.

(1) One
(2) Two
(3) Three
(4) Four

10) Cardiac output is increased by :-

(1) Stimulation of sympathetic nerve
(2) Release of adrenal medullary hormones
(3) Physical exercise
(4) All of these

11) During the process of blood clotting, which of the following correctly represents the sequence of biochemical events leading from injury to the formation of a stable clot, considering the role of thromboplastin, prothrombin, thrombin and fibrinogen ?

(1) Thromboplastin → Prothrombin → Thrombin → Fibrinogen → Fibrin
(2) Thromboplastin → Thrombin → Prothrombin → Fibrinogen → Fibrin

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(3) Fibrinogen → Thromboplastin → Thrombin → Prothrombin → Fibrin

(4) Prothrombin → Thromboplastin → Fibrinogen → Thrombin → Fibrin

12) Identify the incorrect statement about disorders of circulatory system –

(1) High blood pressure may lead to heart diseases but does not affect other vital organs

(2) Atherosclerosis is caused by deposits of calcium, fat, cholesterol and fibrous tissues, making the lumen of arteries narrower

(3) Angina pectoris is an acute chest pain caused when insufficient oxygen is reaching the heart muscles

(4) Heart failure is the state of heart when it is not pumping blood effectively enough to meet the needs of the body

13) Globulin protein of blood plasma is mainly involved in the :-

(1) Clotting

(2) Osmotic balance

(3) Defence mechanism

(4) None

14) The relationship among cardiac output, heart rate and stroke volume is expressed as

(1) Heart rate = Stroke volume × Cardiac output

(2) Cardiac output = Stroke volume × Heart rate

(3) Stroke volume = Cardiac output × Heart rate

(4) Cardiac output = Heart rate ÷ Stroke volume

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15) Most abundant protein of blood plasma is :-

(1) Prothrombin

(2) Albumin

(3) Fibrinogen

(4) Globulin

16) Erythroblastosis foetalis disorder can be seen in which of the following combination :-

	Male Partner	Female Partner
(1)	A^+	AB^+
(2)	AB^+	O^+
(3)	O^+	A^-
(4)	B^-	B^+

(1) 1

(2) 2

(3) 3

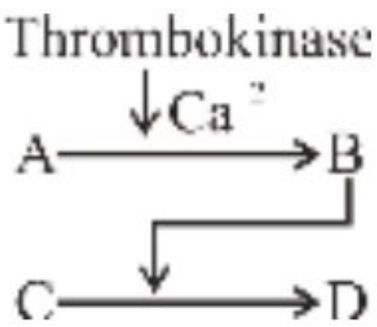
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(4) 4

17) Stoppage of heart beat is called as :-

(1) Angina
(2) Cardiac arrest
(3) Heart block
(4) Heart failure



18) Choose A, B, C and D :-

(1) A → Fibrinogen, B → Prothrombin, C → Fibrin, D → Thrombin
(2) A → Prothrombin, B → Thrombin, C → Fibrinogen, D → Fibrin
(3) A → Fibrin, B → Thrombin, C → Fibrinogen, D → Prothrombin
(4) A → Fibrinogen, B → Thrombin, C → Prothrombin, D → Fibrin

19) Single circuit heart occurs in

(1) Fishes
(2) Frog
(3) Reptiles
(4) Human

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20)

(a) It is colourless fluid containing specialised lymphocytes
(b) It is an important carrier for nutrients, hormones etc
(c) It has clotting capacity
(d) It has no R.B.C., W.B.C. and platelets
(e) Exchange of nutrients, gases etc. between the blood and the cells always occur through this fluid
(f) It has more WBC than blood

How many statements are correct about Lymph?
(1) Two
(2) Three
(3) Four
(4) Five

21) Dead space air is :-

(1) The amount of air remaining in the alveoli
(2) The amount of air left behind in lungs at the end of deep expiration
(3) The amount of air taken in and out

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(4) The air left in the bronchial tree

22) If x is partial pressure of O_2 in deoxygenated blood and y is of CO_2 in oxygenated blood in mm of Hg, than x and y are :-

(1) 40, 40
(2) 95, 40
(3) 40, 45
(4) 45, 40

23) Out of following how many animals respire through gills?
sponge, coelenterates, flat worms, aquatic arthropods, molluscs, fishes, tadpole, reptiles, aves, mammals :-

(1) Three
(2) Four
(3) Six
(4) Eight

24) Additional volume of air, a person can expire by a forcible expiration is :-

(1) IRV (2500-3000 ml)
(2) ERV (1000-1100 ml)
(3) ERV (1500-1600 ml)
(4) EC (1500-1600 ml)

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25)

A disorder in which serious lung damage occur due to proliferation of fibrous tissue is -

(1) Asthma
(2) Emphysema
(3) Occupational respiratory disorder
(4) Bronchitis

26) Effect of CO_2 concentration on dissociation of oxyhaemoglobin is called :

(1) Bohr's effect
(2) Haldane effect
(3) Hamburger effect
(4) Root effect

27) Chloride shift occurs in exchange of?

(1) H^+
(2) K^+
(3) HCO_3^-

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(4) Na^+

28) Match the column-I with column-II and select the correct option.

	Column-I		Column-II
A.	Asthma	(i)	Fibrosis in lungs
B.	Emphysema	(ii)	It is the inflammation of bronchi
C.	Occupational lungs disease	(iii)	Allergy of bronchi and bronchioles
D.	CO_2	(iv)	Carbaminohemoglobin
E	Bronchitis	(v)	Damaging of alveolar wall

Options :-

	A	B	C	D	E
(1)	iii	ii	i	iv	v
(2)	iii	v	i	iv	ii
(3)	ii	v	iv	i	iii
(4)	v	iii	i	ii	iv

(1) 1

(2) 2

(3) 3

(4) 4

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29) Find out the incorrect option :-

(1) High pH and low CO_2 concentration causes left side shifting of oxyHb dissociation curve.

(2) Fetal Hb has more affinity for O_2 molecule than adult Hb

(3) Oxyhaemoglobin dissociation curve is sigmoid in Shape

(4) None of these

30) Arrange the following in the order of increasing volume :
(A) Tidal volume
(B) Residual volume
(C) Inspiratory reserve volume
(D) Vital capacity

(1) A < B < C < D

(2) A < C < B < D

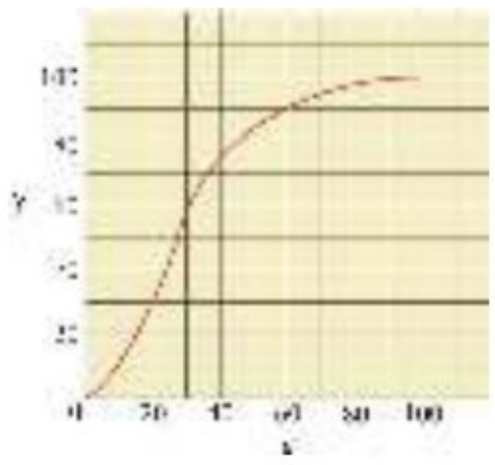
(3) A < D < C < B

(4) A < D < B < C

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31) In the given diagram identify x & y :-



(1) x = percentage saturation of haemoglobin with oxygen y = partial pressure of oxygen (mm Hg)

(2) x = partial pressure of oxygen (mm Hg) y = percentage saturation of haemoglobin with oxygen

(3) x = pressure saturation of haemoglobin with oxygen y = partial pressure of oxygen (mm Hg)

(4) x = Partial pressure of oxygen (cm Hg) y = percentage saturation of haemoglobin with oxygen

32)

Dissociation of oxyhaemoglobin will increase if:-

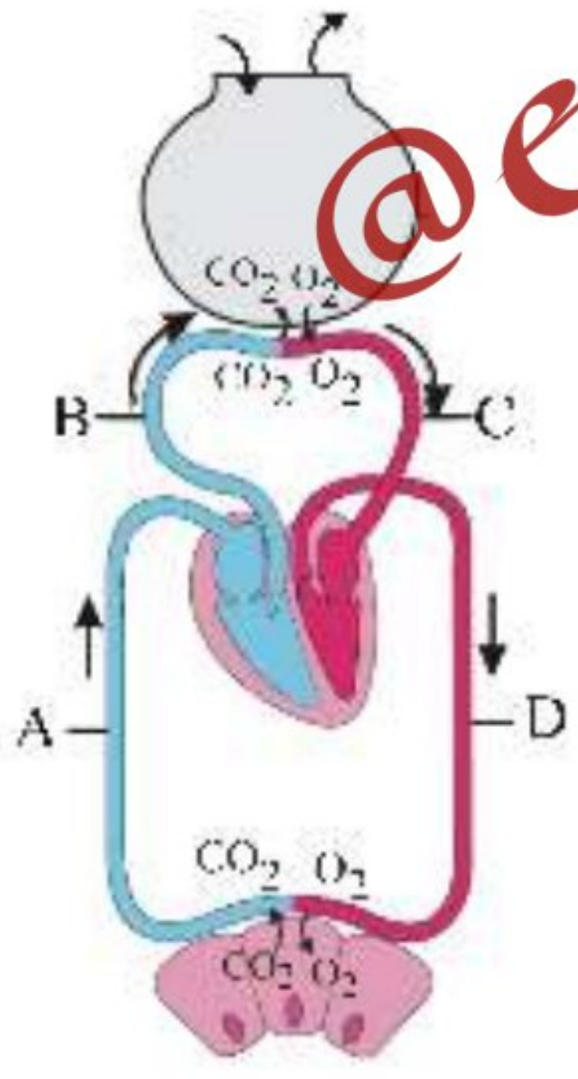
(1) pH of blood falls

(2) pH of blood rises

(3) temperature decreases

(4) pCO_2 decreases

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33) Study the above diagram and select the correct option regarding. Identification and statement :-

(1) A – Systemic vein, pO_2 = 40 mm Hg, pCO_2 = 45 mm Hg

(2) B – Pulmonary Artery, PO_2 = 95-104 mmHg, PCO_2 = 40 mmHg

(3) C – Pulmonary vein, carry oxygenated blood upto systemic organs.

(4) A – Systemic artery, deliver 20 ml of O_2 to tissue fluid

34) Given below is a list of different steps (i–vi) involved in respiration.

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(i) Utilisation of O_2 by the cells for catabolic reactions.
(ii) Transport of gases by the blood.
(iii) Pulmonary ventilation by which atmospheric air is drawn in & CO_2 is released out.
(iv) Release of resultant CO_2
(v) Diffusion of O_2 and CO_2 between blood & tissue
(vi) Diffusion of gases (O_2 and CO_2) across alveolar level.
Select an option which has correct sequence of all the steps :-

(1) iii, vi, ii, v, i, iv
(2) iv, vi, ii, v, i, iii
(3) iii, vi, i, v, ii, iv
(4) iv, ii, v, iii, i, vi

35) Chemosensitive area which is situated adjacent to respiratory rhythm center is highly sensitive to :-

(1) O_2 and CO_2
(2) O_2 and H^+
(3) CO_2 and H^+
(4) O_2, CO_2 and H^+

36) The breathing centre initiates the ventilation in response to :

(1) increase of air pressure
(2) decrease of air pressure
(3) increase of CO_2 in arterial blood
(4) increase of O_2 in arterial blood

37) Which of the following statement is/are true :-
(A) Each haemoglobin molecule can carry a maximum of 4 molecules of O_2
(B) Binding of O_2 with haemoglobin is primarily related to partial pressure of O_2
(C) Every 100 ml blood can deliver around 20 ml of O_2 to the tissue
(D) Oxygen dissociation curve is hyperbola in shape

(1) A , B and D
(2) C and D
(3) A and B
(4) A, B, C and D

38) Which of the following factors favour the formation of Oxyhaemoglobin in lungs ?

(1) $P_{O_2}\uparrow, P_{CO_2}\downarrow$, Temperature$\downarrow$, $H^{\oplus}$concentration$\downarrow$
(2) $P_{O_2}\uparrow, P_{CO_2}\downarrow$, Temperature$\downarrow$, $H^{\oplus}$ concentration$\downarrow$
(3) $P_{O_2}\downarrow, P_{CO_2}\uparrow$,Temperature$\uparrow$, $H^{\oplus}$ concentration$\uparrow$

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(4) $P_{O_2}\uparrow, P_{CO_2}\downarrow$, Temperature$\downarrow$, $H^{\oplus}$ concentration$\downarrow$

39) Read the following four statement (A – D)
(A) Larynx is a cartilagionous box like structure which is helpful in production of sound.
(B) Epiglottis is a white fibrous cartilagionous flap like structure which covers the glottis at the time of swallowing of food.
(C) Trachea is a straight tube like structure which divides at the level of 6th thoracic vertebrae into bronchi in middle of thoracic cavity.
(D) Pharynx is a common path of food and air.
How many of the above statements are **correct**?

(1) Four
(2) One
(3) Two
(4) Three

40) The figure given below shows a small part of human lung where exchange of gases takes place. In which of the options given below, the parts labeled A, B, C or D are correctly identified along with its function ?



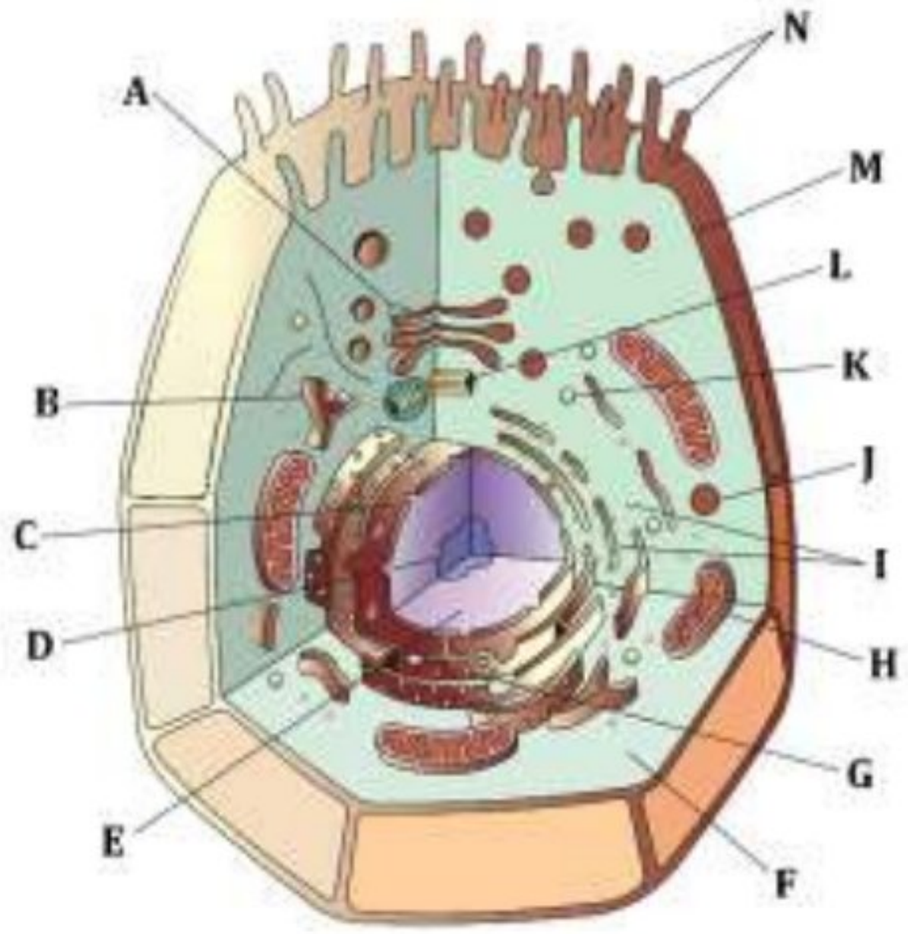
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(1)	B : Red blood cell	transport of CO_2 mainly
(2)	C : Arterial capillary	passes oxygen to tissues
(3)	A : alveolar cavity	main site of exchange of respiratory gases
(4)	D : Capillary wall	exchange of O_2 and CO_2 takes place here

(1) 1
(2) 2
(3) 3
(4) 4

41) Correlate the given features of animal cells (I to VII) with their respective part (A to N)

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I. The structure replicates during mitosis and generates the spindle
II. Major site for synthesis of lipid
III. Power house of the cell
IV. Store house of digestive enzyme
V. Increase the surface are for the absorption of materials
VI. Site of glycolysis
VII. Site for active ribosomal RNA synthesis
The correct option is

(1) I-L, II-G, III-H, IV-J, V-N, VI-F, VII-D
(2) I-M, II-G, III-H, IV-J, V-N, VI-F, VII-D
(3) I-L, II-B, III-H, IV-J, V-N, VI-F, VII-D
(4) I-M, II-A, III-H, IV-J, V-N, VI-F, VII-D

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42) Consider the following five statements about prokaryotic and eukaryotic cells and choose the correct option.
I: The genetic material in prokaryotic cells is a circular, naked DNA, not enclosed by a nuclear envelope.
II: The cell wall of bacteria is made up of peptidoglycan.
III: A well-developed cytoskeleton is present in prokaryotic cells.
IV: Mesosomes are infoldings of the cell membrane in prokaryotes that help in DNA replication and cell wall formation.
V: Eukaryotic cells have 80S ribosomes in the cytoplasm and 70S ribosomes in mitochondria and chloroplasts.

(1) Statements I, II, III, and IV are correct.
(2) Statements I, II, IV, and V are correct.
(3) Statements I, III, and V are correct.
(4) All statements are correct.

43) An unknown cell is observed under an electron microscope. It has a single, circular DNA molecule, 70S ribosomes, and an extensive system of flattened internal membranes containing chlorophyll-like pigments. Which of the following is the MOST likely identity of this cell?

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(1) A typical animal cell mitochondrion

(2) A plant root epidermal cell

(3) An isolated chloroplast from a higher plant

(4) A cyanobacterial cell

44) Which of the following is a UNIQUE characteristic of the nuclear envelope in eukaryotic cells when compared with the plasma membrane?

(1) Presence of embedded proteins in the membrane.

(2) Presence of cholesterol molecules in the lipid bilayer.

(3) Presence of selectively permeable nuclear pores for macromolecule exchange.

(4) Presence of phospholipids in the membrane structure.

45)

Match the following cell types (Column I) with the most appropriate structural feature (Column II):

Column I		Column II	
(P)	Mature mammalian red blood cell	(1)	Enucleate with biconcave shape for gas transport
(Q)	Sieve tube element of phloem	(2)	Contains chloroplasts for photosynthesis
(R)	Bacterial cell	(3)	Lacks a true nucleus but contains a nucleoid
(S)	Parenchyma cell of a leaf mesophyll	(4)	Enucleate but with sieve plate connections for translocation

(1) P-4, Q-1, R-3, S-2

(2) P-1, Q-4, R-3, S-2

(3) P-2, Q-4, R-1, S-3

(4) P-1, Q-2, R-3, S-4

46) **Assertion (A):** Lysosomes are called "suicidal bags" of the cell.
Reason (R): They contain hydrolytic enzymes that are active at an acidic pH and can digest cellular components.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

47) The physiochemical approach to study and understand living organisms is called.

(1) Structural biology

(2) Systems biology

(3) Reductionist biology

(4) Both (2) and (3)

48) Prokaryotes do not possess :-

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(1) Single linear chromosome
(2) Cell wall
(3) Nucleoid
(4) Cell inclusions

49) In a particular plant cell, ribosomes attached to the endoplasmic reticulum are heavily labeled after exposure to a radioactive amino acid. Shortly afterward, dense labeling is also observed in the Golgi cisternae and then in secretory vesicles. This experimental observation MOST strongly supports which function of the rough endoplasmic reticulum (RER)?

(1) Synthesis of all cytosolic enzymes.
(2) Degradation of misfolded proteins via autophagy.
(3) Storage of genetic information for protein synthesis.
(4) Synthesis of proteins destined for secretion or membrane insertion.

50) Transport of molecules through plasma membrane by simple diffusion process takes place for -
(A) Neutral molecules
(B) Polar molecules
(C) Non-polar molecule
(D) Hydrophilic molecule

(1) Only A and B are correct
(2) Only A, C and D are correct
(3) Only A and C are correct
(4) Only A and D are correct

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51) A biologist ground up some plant leaf cells and then centrifuged the mixture to fractionate the organelles. Organelles in one of the heavier fractions could produce ATP in the light, while organelles in the lighter fraction could produce ATP in the dark. The heavier and lighter fractions are most likely to contain, respectively.

(1) Mitochondria and chloroplasts.
(2) Chloroplasts and peroxisomes.
(3) Peroxisomes and chloroplasts.
(4) Chloroplasts and mitochondria.

52) Identify the **incorrect** statement regarding prokaryotic genetic material:

(1) The genetic material is naked and not enveloped by a nuclear membrane.
(2) Many bacteria possess plasmids in addition to genomic DNA.
(3) The genomic DNA in bacteria is typically a single linear chromosome.
(4) Plasmids may confer antibiotic resistance.

53) Statements about the endomembrane system:
I. Includes ER, Golgi complex, lysosomes, and vacuoles.
II. Includes mitochondria, chloroplasts, and peroxisomes.
III. SER is the major site of lipid synthesis.

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IV. RER is abundant in protein-secreting cells.
Which are correct?

(1) I, II, III
(2) I, III, IV
(3) II, III, IV
(4) I and IV only

54) According to the fluid mosaic model (Singer & Nicolson, 1972), membrane "fluidity" refers to:

(1) Free diffusion of water through hydrophobic tails.
(2) Lateral movement of proteins within the lipid bilayer.
(3) Dissolving of peripheral proteins.
(4) Free transport of polar molecules.

55) Identify the incorrectly described chromosome type:

(1) Metacentric – equal arms
(2) Sub-metacentric – unequal arms
(3) Acrocentric – one very short arm
(4) Telocentric – centromere absent

56) In some bacteria, _____ are known to help attach the bacteria to rocks in streams and also to the host tissues.

(1) Cilia
(2) Flagella
(3) Fimbriae
(4) Pili

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57) Which of the following is not true for cilia?

(1) They are hair-like outgrowths of the cell wall.
(2) They cause the movement of either the cell or the surrounding fluid.
(3) Their core is called axoneme.
(4) They are covered with plasma membrane.

58) **Assertion (A) :** Cell is the basic unit of life.
Reason (R) : Anything less than a complete structure of cell does not ensure independent living.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

59)

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Match the column and identify the correct option:

	Column-I		Column-II
(A)	Thylakoids	(I)	Disc-shaped sacs in Golgi apparatus
(B)	Cristae	(II)	Indistinct network of nucleoprotein fibres
(C)	Cisternae	(III)	Flat membranous sacs in stroma
(D)	Chromatin	(IV)	Infolding in mitochondria

(1) A–III, B–IV, C–II, D–I

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-I, C-IV, D-II

60) A cell has a large central vacuole, chloroplasts, and a cell wall made of cellulose. Which additional feature would you MOST confidently expect in this cell type?

(1) Presence of centrioles organizing spindle formation.

(2) Presence of plasmodesmata connecting adjacent cells.

(3) Presence of chitin in the outer most layer.

(4) Presence of numerous contractile vacuoles for osmoregulation.

61) **Assertion:** Meiosis increases the genetic variability in the population of organisms from one generation to the next.
Reason : Meiosis involves pairing of homologous chromosomes and recombination between them.

(1) Assertion and reason both are true and the reason is correct explanation of assertion.

(2) Assertion and reason both are true but reason is not correct explanation of assertion.

(3) Assertion is true but reason is wrong.

(4) Assertion and reason both are wrong.

62) Select the correct option with respect to mitosis.

(1) Chromatids start moving towards opposite poles in telophase

(2) Golgi complex and endoplasmic reticulum are still visible at the end of prophase

(3) Chromosomes move to the spindle equator and get aligned along equatorial plate in metaphase

(4) Chromatids separate but remains in the centre of the cell in anaphase

63) In plant cells, wall formation starts in the **(A)** of the cell and grows **(B)** to meet the existing lateral walls. The formation of the new cell wall begins with the formation of a simple precursor called the **(C)**, which represents the **(D)** between the walls of two adjacent cells. Fill in the blanks with appropriate words.

(1) A → Periphery, B → Centripetally, C → Cell plate, D → Plasmodesmata

(2) A → Center, B → Centrifugally, C → Cell plate, D → Middle lamella

(3) A → Center, B – Centripetally, C → Phragmoplast, D → Primary wall

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(4) A → Periphery, B → Centrifugally, C → Middle lamella, D → Secondary wall

64) Which one of the following statements are correct?
(i) Chromatids start moving towards opposite poles in anaphase.
(ii) Crossing over occurs between non-sister chromatids of homologous chromosomes.
(iii) Crossing over occurs in tetrad stage which occurs in pachytene of prophase I of meiosis I.
(iv) Chromosomes move to the spindle equator and get aligned along equatorial plate in anaphase.
(v) Crossing over is the reciprocal transfer of genes between the non-homologous chromosomes
(vi) M phase represents the phase when actual cell division occurs and I phase represents the phase between two successive M phase.

(1) (i), (ii), (v), (vi)
(2) (ii), (iv), (vi), (vii)
(3) (i), (ii), (iii), (vi)
(4) (ii), (iii), (iv), (v)

65) Match the column I with column II and choose the correct option given below.

Column I		**Column II**	
A.	Centromere splits and chromatids separate apart	I.	Zygotene
B.	Pairing between homologous chromosomes takes place	II.	Point of attachment between homologous chromosome
C.	Chiasmata	III.	Nuclear protein complex that helps in adherence of non-sister chromatids of homologous chromosomes.
D.	Synaptonemal complex	IV.	Anaphase

Codes:
(1) A–III, B–I, C–II, D–IV
(2) A–I, B–IV, C–III, D–II
(3) A–IV, B–I, C–II, D–III
(4) A–II, B–III, C–IV, D–I

66) When the number of chromosomes is already reduced to half in the first reductional division of meiosis, what is the necessity of second meiotic division?

(1) The division is required for the formation of four gametes.
(2) Division ensure equal distribution of haploid chromosomes.
(3) Division ensures equal distribution of genes on chromosomes.
(4) Division is required for the segregation of replicated chromosomes.

67) Which is correct for meiotic metaphase-I?

(1) Bivalents are arranged at equator
(2) Univalents are arranged at equator
(3) Non-homologous chromosomes forms pair

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(4) Spindle fibres are attached at chromomere

68) **Statement I :** Crossing over is non enzymatic process.
Statement II: The final stage of meiotic prophase I is diakinesis

(1) Both statements are correct
(2) Statement I is correct & II is incorrect
(3) Statement I is incorrect & II is correct
(4) Both statements are incorrect

69)

Which of the following statements are wrong regarding synthesis phase?
(i) Period during which DNA synthesis or replication takes place.
(ii) During this time the amount of DNA per cell not changes.
(iii) If the initial amount of DNA is denoted as 2C then it increases to 4C.
(iv) No increase in the chromosome number.
(v) Centriole duplicates in the cytoplasm.
(vi) It is a quiescent stage.

(1) (i), (ii) and (iii)
(2) (ii) and (vi)
(3) (i), (ii) and (iv)
(4) (ii) and (iii)

70) In a meristematic cell, which specific combination best defines the transition from G_1 phase to S phase of the cell cycle?

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(1) Equal distribution of chromosomes to daughter nuclei and onset of cytokinesis.
(2) Completion of spindle formation and disappearance of nuclear envelope.
(3) Sufficient cell size, accumulation of cyclin–CDK complexes, and no DNA damage.
(4) Initiation of chromosome condensation and attachment of microtubules to kinetochores.

71) Which of the following statements are correct?
I. Enzymes decrease activation energy to speed up reactions.
II. Cofactors are required only by oxidoreductases.
III. Enzyme specificity is determined by the 3D shape of the active site.

(1) I and II only
(2) I and III only
(3) II and III only
(4) I, II, and III

72) Which of the following statements are correct?
I. Enzymes remain chemically unchanged after the reaction.
II. Association of coenzyme with apoenzyme is only transient and occur during the course of catalysis.

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III. Competitive inhibitors permanently destroy the enzyme.

(1) I and II only
(2) II and III only
(3) I and III only
(4) I, II and III

73) What happens when the substrate concentration is very low for an enzyme-catalyzed reaction?

(1) Enzyme activity is reduced because fewer enzyme active sites are occupied
(2) Enzyme is denatured immediately
(3) Enzyme gets converted into cofactor
(4) Activation energy increases permanently

74) The correct order of chemical composition of living tissues/cells in term of percentage of the total cellular mass is :-

(1) Nucleic acid > Proteins > H_2O > Carbohydrates > Ions > Lipids
(2) H_2O > Proteins > Nucleic acid > Carbohydrates > Lipids > Ions
(3) H_2O > Proteins > Carbohydrates > Nucleic acid > Lipids > Ions
(4) Lipids > Ions > Carbohydrates > H_2O > Proteins > Nucleic acid

75) **Statement I:** Starch forms linear unbranched secondary structures.
Statement II: Paper made from plant pulp and cotton fibre is rich in starch.

(1) Both statements I and II are correct.
(2) Both statements I and II are incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Statement I is correct, but Statement II is incorrect.

76) **Assertion:** In solution of different pH, the structure of amino acids changes.
Reason: A particular property of amino acids is the ionizable nature of $-NH_2$ and –COOH.

(1) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.
(3) Assertion is true and Reason is false.
(4) Assertion is false and Reason is true.

77) Zinc is a metal ion cofactor for the enzyme:

(1) Peroxidase
(2) Carboxypeptidase
(3) Catalase
(4) Hexokinase

78) In the classification of enzymes, 'Hydrolases' catalyze the:

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(1) Linking together of two compounds
(2) Removal of groups by mechanisms other than hydrolysis
(3) Hydrolysis of ester, ether, or peptide bonds
(4) Inter-conversion of optical isomers

79) The inhibition of succinate dehydrogenase by malonate is a classic example of:

(1) Non-competitive inhibition
(2) Feedback inhibition
(3) Competitive inhibition
(4) Allosteric inhibition

80) Which of the following is an example of a prosthetic group?

(1) NAD
(2) NADP
(3) Haem
(4) Zinc ions

81) Match list-I with list-II.

List-I		**List-II**	
(a)	Tryptophan	(i)	Methyl as a R group
(b)	Lysine	(ii)	Neutral amino acid
(c)	Valine	(iii)	Hydroxy methyl as R group
(d)	Serine	(iv)	Basic amino acid
		(v)	Aromatic amino acid
		(vi)	Acidic amino acid

(1) a–vi, b–v, c–i, d–iii
(2) a–v, b–iv, c–iii, d–i
(3) a–iv, b–vi, c–ii, d–iii
(4) a–v, b–iv, c–ii, d–iii

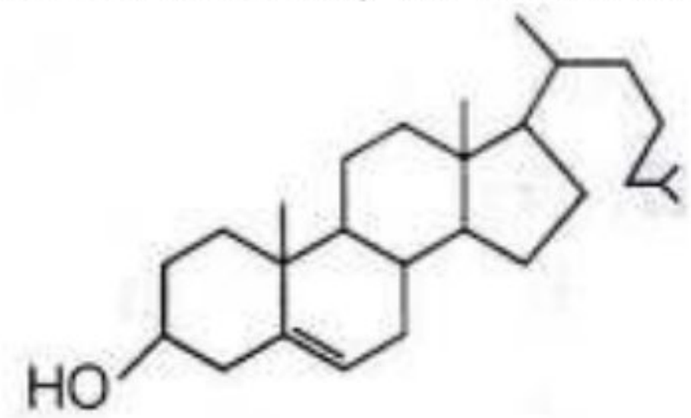
82) Which of the following statements are incorrect?
(a) In proteins, only left handed helices are observed.
(b) Tertiary structure is absolutely necessary for the biological activities of proteins.
(c) In primary structure of protein, the last amino acid is called N-terminal amino acid.
(d) Two subunits of α-type and 4 subunits of β-type together constitute the human haemoglobin (Hb).

(1) a, b & d
(2) a, c & d
(3) b, c & d

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(4) c only

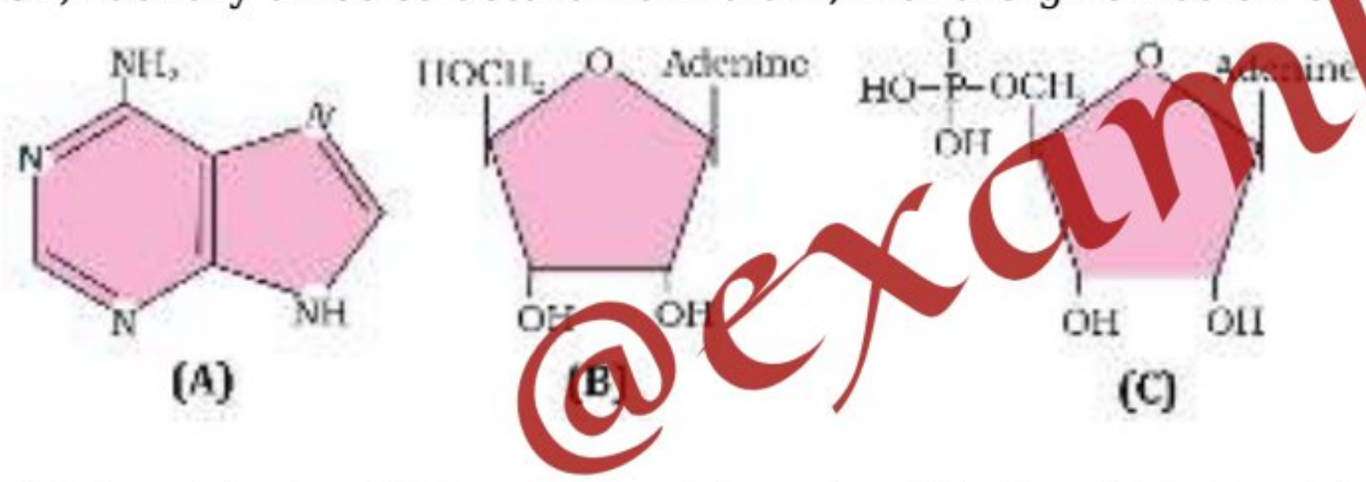
83) Given below is the structure of an animal steroid, which is insoluble in water and chemically unreactive. Study the structure and comment upon the appropriateness of assertion and the reason.



Assertion (A): It is an important component of the membranes of cells, providing stability.
Reason (R): Phospholipid consists of a hydrophilic head containing a phosphate group and two hydrophobic fatty acid tails.

(1) Both A and R are true and R is a correct explanation of A.
(2) Both A and R are true but R is not a correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

84) Identify three structural formula A, B and C given below and choose the correct option.



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(1) A = Adenine (N-base); B = Adenosine (Nucleotide); C = Adenylic acid (Nucleoside)
(2) A = Adenine (N-base); B = Adenosine (Nucleoside); C = Adenylic acid (Nucleotide)
(3) A = Adenosine (Nucleoside); B = Adenylic acid (Nucleotide); C = Deoxyadenylic acid
(4) A = Adenosine (Nucleoside); B = Adenine (N-base); C = Adenylic acid (Nucleotide)

85)

G.N. Ramachandran discovered the triple helical structure of which of the following :
(a) Actin (b) Myosin
(c) Collagen (d) RuBisCO
(e) Insulin

(1) a, b & e
(2) a, b & c
(3) d only
(4) c only

86)

Match List I with List II.

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List-I		List-II	
(a)	Glycerol	(i)	Saturated fatty acid
(b)	Palmitic acid	(ii)	Phospholipid
(c)	Lecithin	(iii)	Simple lipid
(d)	Arachidonic acid	(iv)	Unsaturated fatty acid

(1) a–iv, b–ii, c–i, d–iii

(2) a–iii, b–i, c–ii, d–iv

(3) a–iii, b–iv, c–ii, d–i

(4) a–ii, b–i, c–iii, d–iv

87) Which pair correctly represents a non-reducing sugar and its components?

(1) Sucrose – glucose + fructose

(2) Maltose – glucose + glucose

(3) Lactose – glucose + galactose

(4) Glucose – single unit

88) Exoskeletons of arthropods have a complex polysaccharide called chitin, which is:

(1) A homopolymer of glucose

(2) A heteropolymer of N-acetyl glucosamine

(3) A homopolymer of N-acetyl glucosamine

(4) A storage polysaccharide

89) Match Column I with Column II.

Column-I		Column-II	
(a)	Ligases	(i)	Hydrolysis of C–C, C-halide or P–N bonds
(b)	Transferases	(ii)	S-reduced + S'-oxidised → S-oxidised + S'-reduced
(c)	Lyases	(iii)	Leaving double bonds
(d)	Oxido- reductases	(iv)	Joining of C–O, C–S, C–N, P–O bonds
(e)	Hydrolases	(v)	S–G + S' → S + S'–G

(1) a–i, b–ii, c–iii, d–iv, e–v

(2) a–iii, b–ii, c–iv, d–v, e–i

(3) a–iv, b–v, c–iii, d–ii, e–i

(4) a–iv, b–ii, c–i, d–v, e–iii

90) When comparing Palmitic acid and Arachidonic acid, which statement is true based on the definition of carbon count?

(1) Palmitic acid has 16 carbons excluding the carboxyl carbon.

(2) Arachidonic acid has 20 carbons including the carboxyl carbon.

(3) Palmitic acid has 20 carbons including the carboxyl carbon.

(4) Arachidonic acid has 16 carbons excluding the carboxyl carbon.

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	4	2	2	1	4	2	4	2	2	3	2	2	4	4	1	1	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	1	4	1	1	2	2	3	4	3	4	2	1	3	1	4	2	3	1
Q.	41	42	43	44	45															
A.	3	1	1	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	1	4	3	2	2	3	3	2	3	3	1	4	3	3	4	2	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	4	3	2	2	4	1	4	1	3	3	2	1	4	2	3	2	2	2
Q.	86	87	88	89	90															
A.	4	1	3	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	1	3	1	2	3	1	3	4	1	1	3	2	2	3	2	2	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	2	2	3	1	3	2	4	1	2	1	1	1	3	3	3	4	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	4	3	2	1	3	1	4	3	4	3	2	2	4	3	1	4	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	2	3	3	4	1	3	2	3	2	1	1	2	2	1	2	3	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	2	2	4	2	1	3	3	2										

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SOLUTIONS

PHYSICS

1) **Explaining :** To find self inductance of the solenoid using formula
Concept : This question is based on concept of self inductance
Solution :

$B = \mu_0 \left(\frac{N}{\ell}\right) I$, $\phi = LI = N_B A$

$LI = N\left(\frac{\mu_0 NI}{\ell}\right)A \Rightarrow L = \frac{\mu_0 N^2}{\ell}A$

$L = 4\pi \times 10^{-7} \times \frac{(1200)^2}{48\pi \times 10^{-2}} \times 12 \times 10^{-4}$

$L = 10^{-7} \times (1200)^2 \times 10^{-2} = 144 \times 10^{-5}$ H

or L = 1.44 mH

Final Answer : (3)

2) **Explaining :** To find the emf induced in loop and direction of induced current.
Concept : This question is based on Induced emf.
Solution :

Flux, $\phi = BA$

$= B(\pi R^2)$

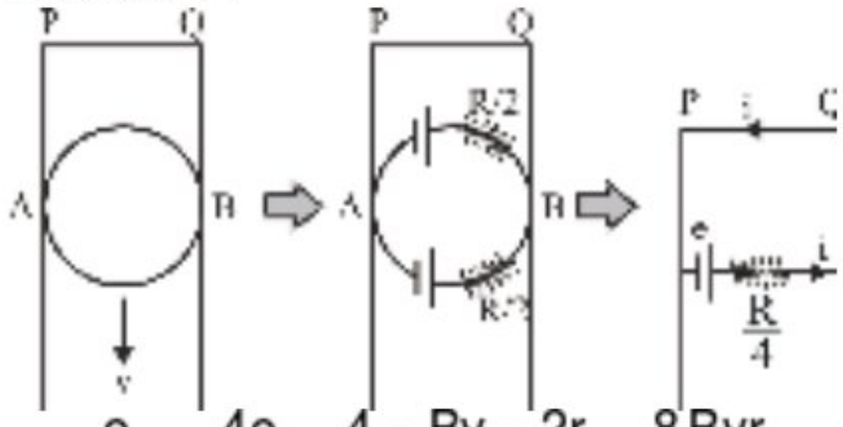
$\phi = \pi B (R_0 + t)^2$

$\varepsilon = \frac{d\phi}{dt} = 2\pi B (R^0 + t)$

Direction of current is anticlockwise

Final Answer : (3)

3) **Explaining :** To find current in section PQ.
Concept : This question is based on motional emf.
Solution :



$i = \frac{e}{R/4} = \frac{4e}{R} = \frac{4 \cdot Bv \cdot 2r}{R} = \frac{8Bvr}{R}$

Final Answer : (4)

4) **Explaining :** To choose the correct statement for transformer.
Concept : This question is based on transformers.
Solution :

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(A) $\frac{V_S}{V_P} = \frac{N_S}{N_P} = 4 \Rightarrow \frac{V_S}{200} = 4 \Rightarrow V_S = 800\ V$

(B) For ideal transformer

$P_{in} = P_{out}$

$V_p.I_P = V_s.I_s$

$200 \times 1 = 800 \times I_S \Rightarrow I_S = 0.25\ A$

(C) 80% Efficiency

$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{V_S I_S}{V_P I_P} \times 100$

$0.8 = \frac{800 \times I_S}{200 \times 1} \Rightarrow I_S = 0.2\ A$

(D) Since $N_S > N_P \Rightarrow$ S.U.T.

Final Answer : (2)

5)

Explaining: The problem asks for the electromotive force (EMF) induced between the center and the rim of a rotating copper disc in a uniform magnetic field.

Concept: This question is based on $\varepsilon = \frac{1}{2}B\omega R^2$.

Solution:

Given values: Radius R=0.1 m. Frequency f=10 rev/s. Magnetic field B=0.1 T.

First, calculate the angular velocity: $\omega = 2\pi f = 2\pi(10) = 20\pi$ rad/s.

Now, calculate the induced EMF: EMF $= \frac{1}{2}B\omega R^2 = \frac{1}{2}(0.1)(20\pi)(0.1)^2$. $=0.5 \times 0.1 \times 20\pi \times 0.01$. Converting this to millivolts: EMF = 10π mV.

Final Answer: (2)

6)

Explaining : A solenoid has turn density 5000 turn/m and cross sectional area 10 cm^2. If a current of 1A flow in it and inside material has relative permeability of 1000, then find energy per unit length of solenoid :-

Concept : This question is based on $u = \frac{1}{2}LI^2$.

Solution : Energy per unit length $= \frac{1}{\ell}\left(\frac{1}{2}LI^2\right)$

$= \frac{1}{\ell}\left[\frac{1}{2}\left(\frac{\mu_0\mu_r N^2 A}{\ell}\right)I^2\right]$

$= \frac{1}{2}\mu_0\mu_r\left(\frac{N}{\ell}\right)^2 AI^2 = \frac{1}{2}\mu_0\mu_r n^2 AI^2$

$= \frac{1}{2} \times 4\pi \times 10^{-7} \times 1000\ (5000)^2\ (10 \times 10^{-4})\ (1)^2$

$= 15.7$ J/m

Final Answer : (1)

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7) **Explain Question:**
Find the terminal velocity required for a rectangular loop to fall through a magnetic field without acceleration.

Concept:
Equilibrium of forces where magnetic force (Upward) equals gravitational force (Downward):
$F_m = mg$

Solution:
The induced emf in the loop is $\epsilon = B\ell v$
The induced current is $I = \frac{B\ell v}{R}$
The upward magnetic force is $F_m = BI\ell = B\left(\frac{B\ell v}{R}\right)\ell$
$F_m = \frac{B^2\ell^2 v}{R}$
For zero acceleration, $F_m = mg$
$\frac{B^2\ell^2 v}{R} = mg$
$v = \frac{mgR}{B^2\ell^2}$
Final answer: Option (4)

8) **Question Explanation:**
Finding the induced current in a rotating small loop.
Concept:
Faraday's Law and Magnetic Flux
Solution:
The magnetic field B at the center of the larger loop of radius x is constant. As the smaller loop rotates, the angle $\theta = \omega t$ between the field and the loop's area vector changes, causing a change in magnetic flux. This change in flux induces an EMF, and subsequently, a current in the loop.
Calculation:
Magnetic Field (B): $B = \frac{\mu_0 i}{2x}$
Magnetic Flux (Φ): $\phi = B \cdot A\cos(\omega t) = \frac{\mu_0 iA}{2x}\cos(\omega t)$
Induced EMF (ϵ): $\varepsilon = -\frac{d\phi}{dt} = \frac{\mu_0 iA\omega}{2x}\sin(\omega t)$
Induced Current (I): $I = \frac{\varepsilon}{R} = \frac{\mu_0 iA\omega}{2xR}\sin(\omega t)$
Final Answer:
Option (2)

9)

Explaining : The energy stored in a coil having inductance L and resistance R is U. Now this coil is divided into four equal parts and connected in parallel to same source. The total energy stored in the combination becomes.

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Concept : This question is based on $U = \frac{1}{2}LI^2$.

Solution : $U = \frac{1}{2}LI^2$

Resistance and Inductance of each part

$$R_1 = \frac{R}{4} \text{ and } L_1 = \frac{L}{4}$$

$$I_1 = \frac{V}{R/4} = \frac{4V}{R}$$

$$U_1 = \frac{1}{2}\frac{L}{4}\left(\frac{4V}{R}\right)^2$$

$$\frac{U}{U_1} = \frac{1}{4}$$

$$\Rightarrow U_1 = 4U$$

Total energy of the combination = $4u_1 = 16u$

Final Answer : (4)

10) **Explaining :** To find the force on charge placed at circumference.

Concept : This question is based on induced electric field.

Solution :

Induced electric field in this loop

$$E(2\pi R) = \pi R^2 \frac{dB}{dt} \text{ (r = R, on the surface)}$$

$$E = \frac{R}{2}\frac{dB}{dt} = 2 \times 10^{-2} \times 0.2 = 4 \times 10^{-3} \text{ V/m}$$

Force applied by this electric field

$$F = qE = 2 \times 4 \times 10^{-3} = 8 \times 10^{-3} \text{ N}$$

Final Answer : (2)

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11)

Explanation: The question asks how the self-inductance L of a solenoid with a fixed number of turns N changes with its length l and area of cross-section A.

Concept: The self-inductance of a solenoid is given by the formula $L = \mu_r \mu_0 n^2 Al$. Here, n is the number of turns per unit length, $n = N/l$. We can substitute this into the formula to see the dependence on N, A, and l.

Solution: The formula for the self-inductance of a solenoid is $L = \mu_r \mu_0 n^2 Al$. We know that $n = N/l$, where N is the total number of turns. Substituting this into the inductance formula:

$$L = \mu_r \mu_0 \left(\frac{N}{l}\right)^2 Al = \frac{\mu_r \mu_0 N^2 A}{l^2} l = \frac{\mu_r \mu_0 N^2 A}{l}$$

From this final expression, we can see that:

A. L is directly proportional to the area A ($L \propto A$).

B. L is inversely proportional to the length l $\left(L \propto \frac{1}{\ell}\right)$.

For the self-inductance to increase, the area A must increase and the length □ must decrease.

Final Answer: (2)

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12)

Explanation: A conducting wire is moving to the right, and the direction of the induced current is shown. The question asks for the direction of the magnetic field.
Concept: This problem can be solved using Fleming's Right-Hand Rule or the vector product formula for magnetic force, $\vec{F}_m = q\left(\vec{v} \times \vec{B}\right)$.
Solution:

A. Let the velocity be $\vec{v} = v\hat{i}$ and the force on a positive charge be in the direction of current, so $\vec{F} = F\hat{j}$.
B. We need t find the direction of $\vec{B}$ such that $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$.
C. $\hat{j} = \hat{i} \times \vec{B}$. The vector that satisfies this is $-\hat{k}$ (the z-axis pointing into the page), since $\hat{i} \times \left(-\hat{k}\right) = -\left(\hat{i} \times \hat{k}\right) = -\left(-\hat{j}\right) = \hat{j}$.

Both methods indicate that the magnetic field is directed perpendicularly into the plane of the paper.
Final Answer: (3) perpendicular to the plane of paper and inwards.

13)

Explanation: A conducting loop is being pulled into a region of uniform magnetic field. The question asks for the direction of the induced current while it is entering and leaving the field.

Concept: This problem is solved using **Lenz's Law**.

Solution:

A. **Entering the magnetic field:** To produce an outward magnetic field, the induced current must flow in the **anticlockwise** direction.
B. **Inside the magnetic field:** According to Faraday's law, if there is no change in flux, no EMF or current is induced.
C. **Leaving the magnetic field:** To oppose decreasing magnetic field induce current must flow **clockwise direction.**

Final Answer: (2)

14)

Explanation: A semicircular wire of radius R is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. A uniform magnetic field of strength B is present. The question asks for the induced EMF between the ends of the wire.

Concept: This is another motional EMF problem, but this time it involves a rotating conductor. The formula for the induced EMF for a rotating conductor is $\varepsilon = \frac{1}{2}B\omega l_{eff}^2$, where l_{eff} is the effective length of the conductor (the straight line distance from the axis of rotation to the other end).

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Solution: The axis of rotation passes through one end of the semicircular wire. The other end is at a distance equal to the diameter of the semicircle, which is 2R. Therefore, the effective length for the rotation is $l_{eff} = 2R$. Using the formula for induced EMF in a rotating conductor:

$\varepsilon = \frac{1}{2}B\omega l_{eff}^2$ $\varepsilon = \frac{1}{2}B\omega(2R)^2 = \frac{1}{2}B\omega(4R^2)$ $\epsilon = 2B\omega R^2$

Final Answer: (2) $2B\omega R^2$

15) **Explanation :-** we are given the magnetic flux ϕ = at (T – t), which varies with respect to time during the time period T, the task is to calculate the heat generated in the loop, assuming the inductance of the coil is negligible.

Concept :- Induced emf

Formula :- The induced e.m.f.ε is related to the rate of change of magnetic flux :
given that ϕ = at(T – t)
induced emf

$$\varepsilon = \frac{d\phi}{dt} = \frac{d}{dt}[at(T-t)] = at(0-1) + a(T-t)$$

= a(T – 2t)
So induced emf is also a function of time
∴ Heat generated in time T is

$$H = \int_0^T \frac{E^2}{R}dt = \frac{a^2}{R}\int_0^T (T-2t)^2 dt = \frac{a^2T^3}{3R}$$

Final Answer : (4)

16) **Explaining :** To determine induced current.

Concept : This question is based on $I_{ind.} = \frac{|e|}{R}$.

Solution :
Emf induced in loop A and loop B.

$$e_A = \frac{d}{dt}\left(B\left(\pi a^2\right)\right) = \pi a^2 k$$

$$e_B = \frac{d}{dt}\left(B\left(\pi b^2\right)\right) = \pi b^2 k$$

$$e_{net} = e_A + e_B = \pi k\left(a^2 + b^2\right)$$

$$R_{net} = (2\pi a)\lambda + (2\pi b)\lambda = 2\pi\lambda(a+b)$$

$$I = \frac{e_{net}}{R_{net}} = \frac{\pi k\left(a^2+b^2\right)}{2\pi\lambda(a+b)} = \frac{k}{2\lambda}\frac{\left(a^2+b^2\right)}{(a+b)}.$$

Final Answer : (4)

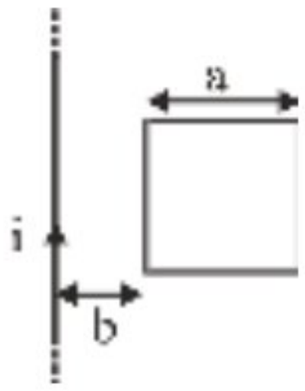
17)

Explaining :

Find the mutual inductances between the straight wire and the square loop shown in figure.

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Concept : This question is based on $M = \frac{\phi}{I}$.

Solution : $d\phi = BdA$

$$d\phi = \frac{\mu_0 I}{2\pi x} a dx$$

$$\phi = \frac{\mu_0 Ia}{2\pi} \int_b^{b+a} \frac{dx}{x}$$

$$\phi = \frac{\mu_0 Ia}{2\pi} [\ell n x]_b^{b+a}$$

$$\phi = \frac{\mu_0 Ia}{2\pi} \left[\ell n \left(\frac{b+a}{b}\right)\right]$$

$$M = \frac{\phi}{I} = \frac{\mu_0}{4\pi}.2a\ \ell n \frac{a+b}{b}$$

$$M = \frac{\mu_0 a}{2\pi}.\ell n \left(1 + \frac{a}{b}\right)$$

Final Answer : (1)

18) **Explaining :** To choose the correct statement.

Concept : This question is based on R–L circuit.

Solution :

Saturation current is same,

$$i_{0_A} = I_{0_B}$$

$$\left(\frac{E}{R}\right)_A = \left(\frac{E}{R}\right)_B$$

$\because E_A = E_B$

$R_A = R_B$

$\tau_B > \tau_A$

$$\frac{L_B}{R} > \frac{L_A}{R}$$

$L_B > L_A$

$L \uparrow$

Final Answer : (1)

19) **Explaining :** Find ratio of self inductances.

Concept : This question is based on $L \propto N^2 r$.

Solution : $L = \frac{\mu_0 \pi N^2 r}{2}$

$L \propto N^2 r$

$$\frac{L_1}{L_2} = \frac{(3n)^2 \times r}{n^2 \times 3r} = \frac{3}{1}$$

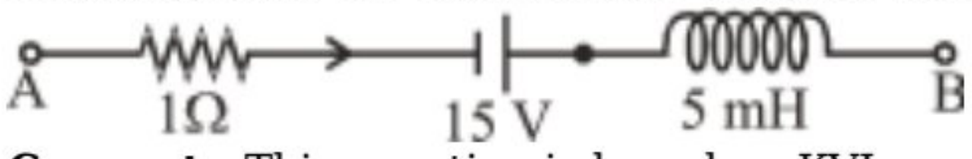
Final Answer : (2)

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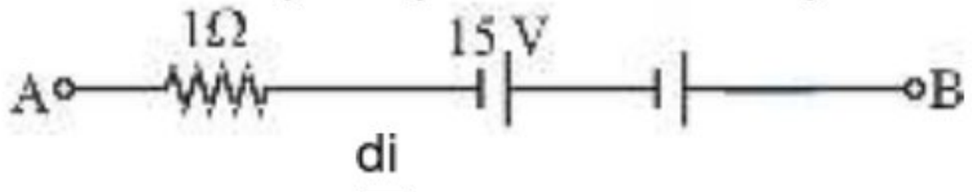
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20) **Explaining :** The network shown in the figure is a part of a complete a circuit. If at a certain instant the current i is 5 A and is decreasing at the rate of 10^3 A/s then $V_B - V_A$ is :-



Concept : This question is based on KVL.

Solution : By using Kirchhoff's voltage law



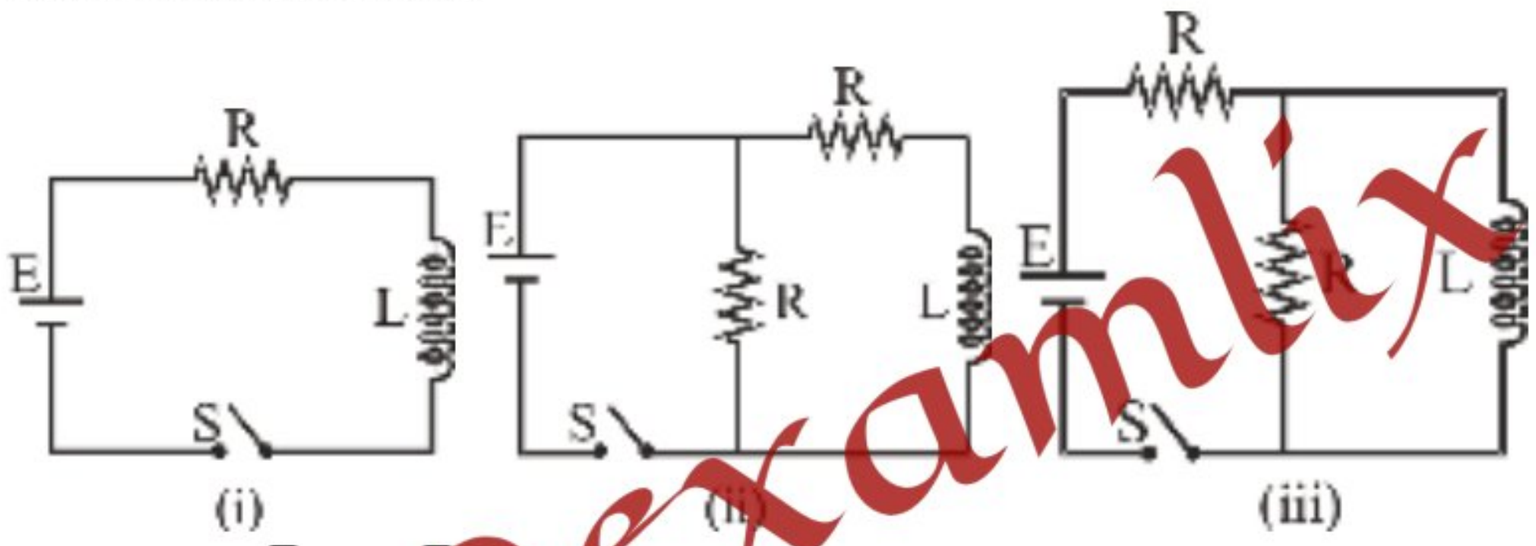
$V_A - i R + E + L\frac{di}{dt} = V_B \Rightarrow V_B - V_A = 15$ volt.

Final Answer : (3)

21) **Explaining :** Find current at t = 0.

Concept : This question is based on $I = \frac{E}{R}$.

Solution : At t = 0 current through L is zero so it acts as open circuit. The given figures can be redrawn as follows.



$i_1 = 0;\ i_2 = \frac{E}{R} i_3 = \frac{E}{2R}$

Hence $i_2 > i_3 > i_1$.

Final Answer : (2)

22) **Explaining :** The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m^2 acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1μV, is :-

Concept : This question is based on $e = \frac{d\phi}{dt}$.

Solution : $e = \frac{d\phi}{dt} = \frac{d}{dt}(\pi r^2 B) = 2\pi r B \frac{dr}{dt}$

$\Rightarrow r = \frac{e}{\left(2\pi B \frac{dr}{dt}\right)}$

$\Rightarrow r = \frac{10^{-6}}{2\pi \times 10^{-3} \times 10^{-2}} = \frac{5}{\pi}$cm

Final Answer : (4)

23) **Explaining :** Find M.

Concept : This question is based on $\varepsilon_2 = M\frac{dI_1}{dt}$.

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Solution : $\varepsilon_2 = M\frac{dI_1}{dt}$

$I_2R_2 = M\frac{dI_1}{dt}$

$\frac{dI_1}{dt} = \frac{I_2R_2}{M} = \frac{0.4 \times 5}{0.5} =$ 4.0 A/s

Final Answer : (1)

24) **Explaining :** Given I - t graph find V - T graph.

Concept : This question is based on $\varepsilon = -L\frac{dI}{dt}$.

Solution : $\varepsilon = -L\frac{dI}{dt}$

0 to $\frac{T}{2}$, $\frac{dI}{dt}$ = +ve constant ⇒ ε = –ve constant

$\frac{T}{2}$ to T, $\frac{dI}{dt}$ = –ve constant ⇒ ε = +ve constant.

but best option is (4)

Final Answer : (4)

25)

Question Explanation: Two statements are given related to eddy currents and their physical effect. We must judge their correctness.

Concept: Eddy currents (Lenz's law).

Solution

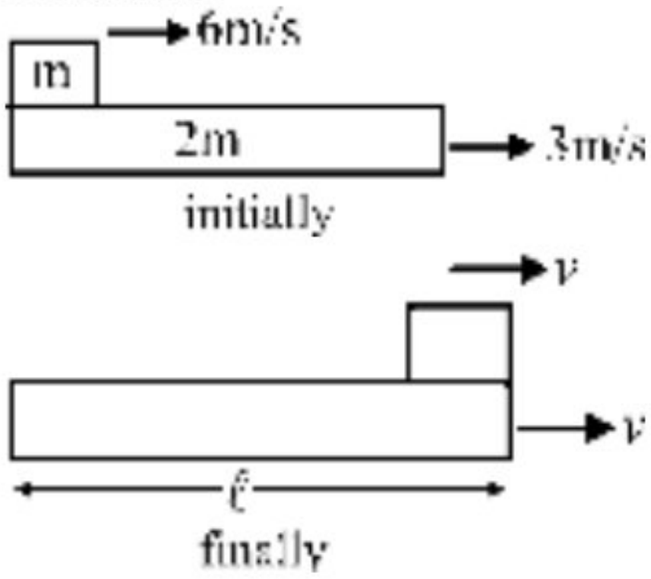
A. Statement I: Eddy currents are used in electromagnetic induction brakes — True
B. Eddy currents induced in a conductor produce a retarding force, which is used for braking.
C. Statement II: Eddy currents always oppose the relative motion — True
D. According to Lenz's law, the induced currents oppose the change or motion producing them.

Final Answer : Option (1)

26) **Explaining :** To find the velocity of plank

Concept : This question is based on concept of centre of mass.

Solution:



$p_i = p_f$

$m \times 6 + 2m \times 3 = (m + 2m)V$

$V = 4m/s$

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Final Answer: The correct option is (1).

27) **Explaining :** To find extension of spring when the power due to spring force on the block is maximum.
Concept : This question is based on concept of potential energy of spring.
Solution : When extentiaon is x

$$\frac{1}{2}k(x_0^2 - x^2) = \frac{1}{2}mv^2$$

$$P = \vec{F}.\vec{V} = kxV$$

$$\frac{dp}{dx} = 0 \Rightarrow x = \frac{x_0}{\sqrt{2}}$$

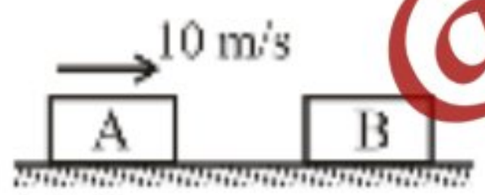
Final Answer : (2)

28) **Explaining :** To find magnitude of the displacement of the boat.
Concept : This question is based on concept of centre of mass.
Solution : Let mass of boat = M; Mass of man = m
displacement of boat

$$= \frac{m}{M+m}L = \frac{50}{450+50} \times 10 = 1m$$

L = 1m travelled distance
Final Answer : (2)

29) **Explaining :** To find speed of the block B just after the collision.
Concept : This question is based on concept of collision.
Solution :



A & B :



$10 = v_1 + v_2 \quad ....(1)$

$$e = \frac{v_2 - v_1}{10} = 0.5 \Rightarrow v_2 - v_1 = 5 \quad ....(2)$$

$\therefore v_2 = 7.5$ m/s
Final Answer : (3)

30) **Explaining :** To find the value of a.
Concept : This question is based on concept of oblique collision.
Solution : After 2 sec, they both will have same position due to collision

$$\therefore \vec{S}_1 = \vec{v}_1 t$$

$$\vec{r}_f - (3\hat{i} + 5\hat{j}) = (4\hat{i} + 3\hat{j}) \times 2$$

$$\vec{r}_f = 11\hat{i} + 11\hat{j}$$

$$\vec{S}_2 = \vec{v}_2 t$$

$$\vec{r}_f - (-5\hat{i} - 3\hat{j}) = (a\hat{i} + 7\hat{j}) \times 2$$

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$\vec{r}_f = (2a-5)\hat{i} + 11\hat{j}$

$11\hat{i} + 11\hat{j} = (2a-5)\hat{i} + 11\hat{j}$

$2a - 5 = 11$

$a = \frac{16}{2} = 8$

Final Answer : (4)

31)

Explaining : To find the speed of bullet.

Concept : This question is based on concept of conservation of linear momentum.

Solution :

By using equation $(V')^2 = u^2 + 2gh = \sqrt{2 \times 10 \times 5}$

$= V' = 10\,m/s$

By principle of conservation of linear momentum

$(0.1)V + 0 = (0.1 + 0.4)V'$

$= V' = \frac{V}{5} \Rightarrow V = 5V'$

= we get V = 50 m/s

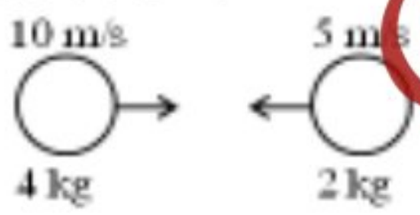
Final Answer : (3)

32)

Explaining : To find coefficient of restitution

Concept : This question is based on concept of coefficient of restitution.

Solution :



$(4 \times 10) - (2 \times 5) = 4V_A + 2 \times 7$

$\Rightarrow \quad V_A = 4$ m/s

Now $e = \frac{V_{sep}}{V_{app}} = \frac{7-4}{10+5}$

$\Rightarrow \quad e = \frac{1}{5}$

Final Answer : (4)

33) **Explaining :** To find the kinetic energy of the 6kg mass.

Concept : This question is based on concept of bomb explosion.

Solution :

$p^2 = 2m_1k_1$

$p^2 = 2m_2k_2$

$\Rightarrow m_1k_1 = m_2k_2$

$k_2 = \frac{m_1k_1}{m_2}$

$k_2 = \frac{3 \times 216}{6} = 108$ J

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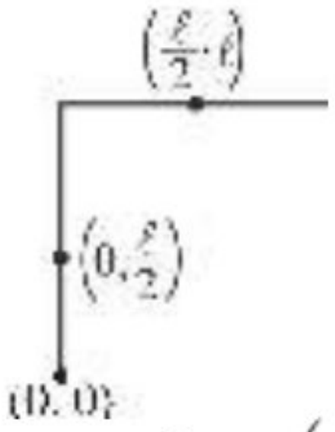
Final Answer : (2)

34) **Explaining :** A 1 kg stationary bomb explodes into three parts with mass ratio 4:4:7. The two smaller parts move perpendicularly each with velocity 70 m/s. Find the velocity of the larger part.
Concept : This question is based on Conservation of momentum in vector form: total momentum before explosion is zero, so vector sum of momenta after explosion is zero.
Solution :

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = 0$$
$$\vec{p}_3 = -(\vec{p}_1 + \vec{p}_2)$$
$$7m = 4m \times 70\sqrt{2}$$
$$m = 40\sqrt{2}$$

Final Answer : (1)

35) **Explain the question :** As shown in figure find co-ordinates of C.O.M.
Concept : This question is based on center of mass.
Solution :



$$X_{cm} = \frac{0 + m\left(+\frac{\ell}{2}\right)}{2m}$$
$$X_{cm} = \frac{\ell}{4}$$
$$Y_{cm} = \frac{m \times \frac{\ell}{2} + m \times \ell}{2m}$$
$$Y_{cm} = \frac{3\ell}{4}$$

Final answer : 3

36) **Explaining :** To find the value of x.
Concept : This question is based on concept of conservation of momentum.
Solution :
Using conservation of momentum

$$(0.1)\left(3\hat{i}\right) + (0.1)(5\hat{j}) = (0.1)4(\hat{i} + \hat{j}) + (0.1)\vec{V}_B$$
$$\vec{V}_B = -\hat{i} + \hat{j}$$
$$\left|\vec{V}_B\right| = \sqrt{2}$$

Kinetic energy of B is $KE = \frac{1}{2}mV_B^2$

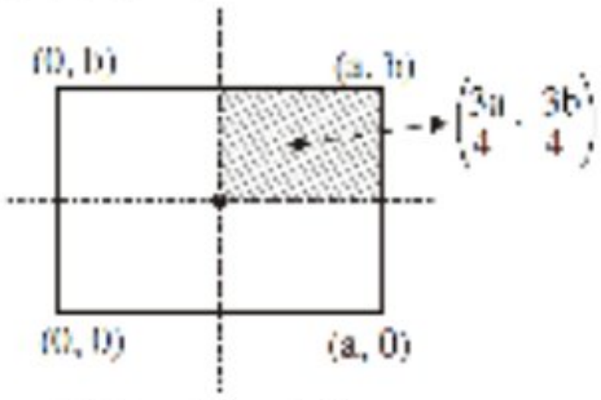
$$KE = \frac{1}{10}J$$

Final Answer : (1)

37) **Explaining :** To find the coordinates of the centre of mass of the remaining portion.

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Concept : This question is based on concept of centre of mass.
Solution :

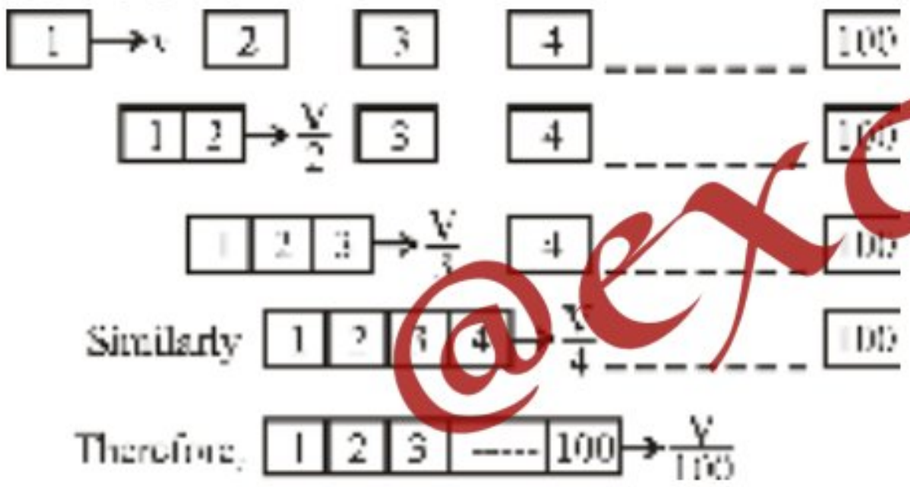


$$x = \frac{M\frac{a}{2} - \frac{M}{4} \times \frac{3a}{4}}{M - \frac{M}{4}} = \frac{\frac{a}{2} - \frac{3a}{16}}{\frac{3}{4}} = \frac{\frac{5a}{16}}{\frac{3}{4}} = \frac{5a}{12}$$

$$y = \frac{M\frac{b}{2} - \frac{M}{4} \times \frac{3b}{4}}{M - \frac{M}{4}} = \frac{5b}{12}$$

Final Answer : (4)

38) **Explain the question :** For the system of hundred identical blocks, we need to find the final velocity of the set of hundred stucked blocks, from data given.
Concept : This question is based on head-on collisions (perfectly inelastic)
Solution :



Final answer : 2

39) **Explanation of Question** : A 1kg particle bounces off the ground with specific incoming and outgoing velocities and angles. Calculate the impulse.
Concept Used: Impulse-Momentum Theorem- Impulse is the change in linear momentum, calculated using vector subtraction of the velocities
Solution:

$$J = m(\vec{v}_2 - \vec{v}_1)$$

$$J = m\left[\left(15\hat{i} + 15\hat{j}\right) - \left(25 \times \frac{3}{5}\hat{i} - 25 \times \frac{4}{5}\hat{j}\right)\right]$$

$$= m\left[15\hat{j} + 20\hat{j}\right] = (1 \times 35)\hat{j}$$

Final Answer : (3)

40) **Explaining :** To match the List–I and II.
Concept : This question is based on concept of centre of mass.
Solution :

(I) zero

(II) $$O = \frac{90(x+2) + 120(x-2) + 90x}{90 + 120 + 90} = 0.2$$

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(III) $x + 2 = 2.2m$ (Right)
(IV) $x - 2 = -1.8m$ (Left)
Final Answer : (1)

41) **Explaining :** To find coefficient of restitution e.
Concept : This question is based on concept of coefficient of restitution.
Solution :
$3 \times 7 = 3v_1 + 5\ v_2$
$2v_2 = 3v_1$
$v_1 = 2m/s; \quad v_2 = 3\ m/s$
$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{1}{7}$
Final Answer : (3)

42) **Explaining :** To find the position of centre of mass of rod in meters.
Concept : This question is based on concept of centre of mass of continuous system.
Solution :

$$x_{cm} = \frac{\int x dm}{\int dm} = \frac{\int_0^6 x\left(kx^4 dx\right)}{\int_0^6 \left(kx^4 dx\right)} = \frac{\int_0^6 x^5 dx}{\int_0^6 x^4 dx} = \frac{\left(\frac{x^6}{6}\right)_0^6}{\left(\frac{x^5}{5}\right)_0^6} = 5m$$

Final Answer : (1)

43) **Explaining :** To find the distance of centre of mass of the particles from the fixed point.
Concept : This question is based on concept of centre of mass of discrete system.
Solution :

$$\ell_{cm} = \frac{\sum_{i-1}^{n} m_i \ell_i}{\sum_{i=1}^{n} m_i} = \frac{m\ell \sum_{i-1}^{n} n^2}{m\Sigma n}$$

$$= \frac{n(n+1)(2n+1)}{6 \times \frac{n(n+1)}{2}} = \left(\frac{2n+1}{3}\right)\ell$$

Final Answer : (1)

44) **Explaining :** To find direction of centre of masss.
Concept : This question is based on concept of bomb explosion.
Solution : Internal forces does not change the path of centre of mass.
Final Answer : (4)

45) **EXPLANATION :**
We must identify the false statement among four options related to momentum andcollisions.
CONCEPT:
Momentum vs kinetic energy in collisions — momentum is always conserved for an

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isolatedsystem; kinetic energy need not be.
SOLUTION:
(1) The magnitude of momentum of a heavy object is greater than that of a light object moving at the samespeed. Momentum p = mv. For the same speed v, if mass m is larger, momentum p is larger. This statement is true.
(2) In a perfectly inelastic collision, all the initial kinetic energy of the colliding bodies is dissipated.
A perfectly inelastic collision is one where objects stick together, and the kinetic energy loss is maximum.However, it's not "all" the kinetic energy unless the final velocity of the combined mass is zero. If the finalkinetic energy is not zero, then not all of the initial kinetic energy is dissipated. This statement is false.
(3) The momentum of a system of colliding bodies may be conserved even though the total mechanicalenergy may not.
Momentum is conserved in all types of collisions (elastic, inelastic, perfectly inelastic) as long as noexternal forces act on the system. Mechanical energy, however, is only conserved in elastic collisions. This statement is true.
(4) The velocity of the center of mass of a system is the system's net momentum divided by its total mass. This statement is true.
Answer : (2)

CHEMISTRY

46)

Explaining : The occurrence of a reaction is impossible if :-
Concept : This question is based on $\Delta G = \Delta H - T\Delta S$
Solution : When ΔH is positive and ΔS is negative, then $\Delta G > 0$
Final Answer : (1)

47) **Explaining :** Internal energy change during a reversible isothermal expansion of an ideal gas is :-
Concept : This question is based on Internal energy.
Solution :
Internal Energy change = ?
In reversible isothermal expansion
In isothermal, temperature remains constant so internal energy change is zero
Final Answer : (3)

48)

Explaining : Difference in ΔH and ΔE
Concept : This question is based on $\Delta H = \Delta E + \Delta n_g RT$
Solution :
$CH_{4\,(g)} + 2O_{2\,(g)} \rightarrow CO_{2\,(g)} + 2H_2O_{(\square)}$
$\Delta H - \Delta E = \Delta n_g RT$
$= (-2)\,(2)\,(298)$ cal
Final Answer : (1)

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49)

Explaining : Relation between ΔH and ΔE
Concept : This question is based on $\Delta H = \Delta E + \Delta n_g RT$
Solution :

$$C_4H_{10(g)} + \frac{13}{2}O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(g)}$$

$\Delta n_g = +1.5 \qquad \therefore \Delta H > \Delta E$
Final Answer : (4)

50)

Explaining : Internal energy
Concept : This question is based on internal energy
Solution : Internal energy of an ideal Gas depends on temperature.
Final Answer : (3)

51)

Explaining : Entropy change
Concept : This question is based on entropy change during phase transition.
Solution :

$$\Delta S = \frac{\Delta H}{T}$$

$\Delta H = 540$ cal/gm
$= 540 \times 18$ cal/mol.

$$\Delta S = \frac{540 \times 18}{373}$$

$= 26.05$ cal/K mol
Final Answer : (2)

52)

Explaining : Change in entropy
Concept : This question is based on entropy change for a thermodynamic process
Solution :
Isothermal Reversible

$$\Delta S = 2.303\ nR \log\left(\frac{V_2}{V_1}\right)$$

$$= 2.303 \times 5 \times 8.314 \times \log\frac{80}{8}$$

$= 95.73\ JK^{-1}$
Final Answer : (2)

53)

Explaining : Change in internal energy
Concept : This question is based on FLOT

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Solution :
Work done by system W = –8J
Heat is supplied q = 40J
From 1st Law-
$\Delta U = q + W$
$\Delta U = 40–8$
$\Delta U = 32J$
Final Answer : (3)

54)

Explaining : Relation between ΔH and ΔE
Concept : This question is based on $\Delta H = \Delta E + \Delta n_g RT$
Solution :

$$Ag_2O(s) \rightarrow 2Ag(s) + \frac{1}{2} O_2(g)$$

$$\Delta n_g = \frac{1}{2}$$

$\Delta H = \Delta E + \Delta n_g RT$
$\Delta H = \Delta E + (0.5)RT$
$\Delta H > \Delta E$
Final Answer : (3)

55)

Explaining : Work done is
Concept : This question is based on work done
Solution :
n = 2 mol
Vaccum, (in vaccum work done is zero)
W = 0
Final Answer : (2)

56)

Explaining : ΔS will be maximum
Concept : This question is based on entropy
Solution :
Gaseous moles are increasing, so entropy increases.
(ii) gaseous moles are increasing
(i), (iii), (iv) gaseous moles are same.
Final Answer : (3)

57)

Explaining : ΔH – ΔE
Concept : This question is based on $\Delta H = \Delta E + \Delta n_g RT$
Solution :

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$\Delta H - \Delta E = \Delta n_g RT$

$= \left(\frac{1}{2}\right)(0.002 \times 1\,000\text{cal})\,T$

Final Answer : (3)

58)

Explaining : Change in enthalpy
Concept : This question is based on $\Delta H = \Delta E + \Delta(PV)$
Solution :
$\Delta H = \Delta U + \Delta(PV)$
$\Delta H = \Delta U + (P_2V_2 - P_1V_1)$
$\Delta H = 40 + (20-3)$
$\Delta H = 57$ L-atm
Final Answer : (1)

59)

Explaining : Work done
Concept : This question is based on work done in reversible isothermal process.

Solution : $w = -2.303\ nRT \log\left(\frac{V_2}{V_1}\right)$
Final Answer : (4)

60)

Explaining : Entropy chnage
Concept : This question is based on entropy change for a the thermodynamics process.
Solution :

$\Delta S = nc_v \ln \frac{T_1}{T_2} + nR \ln \frac{p_1}{p_2}$

For isothermal $T_1 = T_2$

$\Delta S = nR \ln \frac{p_i}{p_f}$

Final Answer : (3)

61)

Explaining : Work done
Concept : This question is based on work done
Solution :
For reversible expansion
$n = 1$ mol
$V_1 = 10L, \qquad V_2 = 20L$
For isothermal reversible expansion
$1\ J = 10^7 erg$

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$$W = -2.303\ nRT \log_{10}\left(\frac{V_2}{V_1}\right)$$

$$W = -2.303 \times 1 \times 8.314 \times 298 \times 10^7 \log \frac{2}{1}$$

Final Answer : (3)

62)

Explaining : Enthalpy of vaporisation
Concept : This question is based on enthalpy of reaction
Solution :

$H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$; ΔH : + 285.77 kJ/mol

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; ΔH = –241.84 kJ/mol

(Adding) H_2O (l) → $H_2O(g)$; ΔH: +43.93 kJ/mol

Final Answer : (4)

63)

Explaining : ΔG°
Concept : This question is based on relation between ΔG° and K_{eq}
Solution :

$\Delta G^\circ = -2.303\ RT \log K_p$

$= -2.303 \times 8.314 \times 298 \log (4.42 \times 10^4)$

$= -26.5\ kJ\ mol^{-1}$

Final Answer : (2)

64)

Explaining : Enthalpy of combustion
Concept : This question is based on heat of combustion
Solution : The reaction of combustion is :

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_20(l)$

$\Delta H_{reaction} = [2\Delta_f H^\circ_{CO_2} + 2\Delta_f H^\circ_{H_2O}] - [\Delta_f H^\circ_{C_2H_4} + 3\Delta_f H^\circ_{O_2}]$

$= [2(-394) + 2(-286)] - [52 + 0]$

$= -1360 - 52 = -1412\ kJ\ mol^{-1}$

Final Answer : (2)

65)

Explaining : Work done
Concept : This question is based on work done in reversible isothermal process
Solution :

$$w = -2.303nRT \log \frac{V_2}{V_1}$$

$$w = -2.303 \times 5 \times 8.314 \times 300 \log \frac{60}{6}$$

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w = –28.72 kJ
Final Answer : (4)

66)

Explaining : Theory of spontaneous and non-spontaneous
Concept : This question is based on theory of spontaneous and non-spontaneous
Solution :
ΔH = –ve (Reaction is the combustion)
Δn_g = (16+18) – (2 + 25)
= +ve
$\therefore \Delta S$ = +ve
ΔG = –ve (Reaction is spontaneous).
Final Answer : (1)

67)

Explaining : Equilibrium constant
Concept : This question is based on relation between $\Delta G°$ and K_{eq}
Solution :
$\Delta G° = -2.303\ RT \log K$
When K = 1, $\Delta G° = 0$
Final Answer : (3)

68)

Explaining : Relation between ΔH and ΔU
Concept : This question is based on $\Delta H = \Delta U + \Delta(PV)$
Solution :
$\Delta H = \Delta U + P\Delta V$
$\Delta U = \Delta H - P\Delta V$
Final Answer : (4)

69)

Explaining : The correct relation between standard Gibb's energy change ($\Delta G°$) and Equilibrium constant (K_p) is :
Concept : This question is based on relation between $\Delta G°$ and K_{eq}
Solution :
Vant Hoff equation
$\Delta G^0 = -RT\log_e K_p$
$-\Delta G^0 = RT\ln K_p$
Final Answer : (3)

70)

Explaining : Melting point of the liquid is :-
Concept : This question is based on entropy change during phase transition

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Solution : $\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T_{mp}}$

$5.26 = \frac{1.435 \times 1000}{T_{mp}}$

$T_{mp} = \frac{1435}{5.26} = 273K$, i.e. 0°C

Final Answer : (2)

71)

Explaining : Spontaneous

Concept : This question is based on theory of spontaneous and non-spontaneous process.

Solution :

$\Delta S_{total} = \Delta S_{system} + \Delta S_{surrounding}$

= +ve for spontaneous process

Final Answer : (2)

72)

Explaining : Find ΔS

Concept : This question is based on $\Delta G = \Delta H - T\Delta S$

Solution : $\Delta G = 0$,

$\therefore \Delta H - T\Delta S = 0$

$\Delta S = \frac{\Delta H}{T} = \frac{6000}{273} = 21.9\ J\ K^{-1}mol^{-1}$

Final Answer : (4)

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73)

Explaining : $\Delta G° = ?$

Concept : This question is based on $\Delta G = \Delta H - T\Delta S$

Solution :

$\Delta G° = \Delta H° - T\Delta S°$

$= -382.64 \times 1000 - 298 \times (-145.6)$

$= -33925\ J = -339.25\ kJ$

Final Answer : (1)

74)

Explaining : What is ΔH

Concept : This question is based on enthalpy of reaction

Solution : Subtracting equation 1st from 2nd multiplied by 3,

$2Fe(s) + \frac{3}{2}O_2(g) \rightarrow Fe_2O_3(s)$ $(\Delta H = -193.4\ kJ)$ ...(i)

$3Mg(s) + \frac{3}{2}O_2(g) \rightarrow 3MgO(s)$ $(\Delta H = -420.6\ kJ)$ ...(ii)

Subtracting eq. (i) from (ii),

$3Mg(s) + Fe_2O_3 \rightarrow 3MgO + 2Fe$; $[\Delta H = -420.6 - (-193.4) = -227.2\ kJ]$

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Final Answer : (4)

75)

Explaining : ΔH = ?
Concept : This question is based on enthalpy of reaction
Solution :

$$2FeO(s) \longrightarrow 2Fe(s) + O_2(g); \quad \Delta H^\circ = +544\ kJ$$
$$2Fe_2O_3(s) \longrightarrow 4Fe(s) + 3O_2(g); \quad \Delta H^\circ = +1648\ kJ$$
$$6Fe(s) + 4O_2(g) \longrightarrow 2Fe_3O_4(s); \quad \Delta H^\circ = -2 \times 1118.4\ kJ$$
$$\text{On adding, } 2FeO(s) + 2Fe_2O_3(s) \longrightarrow 2Fe_3O_4(s); \quad \Delta H^\circ = -44.4 kJ$$

$\therefore FeO(s) + Fe_2O_3(s) \rightarrow Fe_3O_4(s);\ \Delta H = -22.2\ kJ$
Final Answer : (1)

76)

Explaining : Average bond energy of (O–H)
Concept : This question is based on bond energy
Solution : $2(H–H) + O=O \rightarrow 2(H–O–H)$

$\Delta H = \Sigma(BE)_{reactants} - \Sigma(BE)_{products}$

$-571 = [2 \times 435 + 498] - 4 \times (BE)_{O-H}$

$$(BE)_{O-H} = \frac{2 \times 435 + 498 + 571}{4} \approx 484\ kJ$$

Final Answer : (3)

77)

Explaining : ΔH
Concept : This question is based on bond energy
Solution :

$\Delta H_{reaction} = \Sigma(BE)_{reactants} - \Sigma(BE)_{products}$

$= [(BE)_{c=c} + 4(BE)_{C-H} + (BE)_{Cl-Cl}] - [4(BE)_{C-H} + 2(BE)_{C-Cl} + (BE)_{C-C}]$

$= [(BE)_{C=C} + (BE)_{Cl-Cl}] - [2(BE)_{C-Cl} + (BE)_{C-C}]$

$= [612 + 243] - [2 \times 331 + 347] = -154\ kJ$

Final Answer : (3)

78)

Explaining : Oxidation state of chlorine
Concept : This question is based on oxidation number
Solution :

$HClO_4$: $+1 + x - 8 = 0,\ x = +7$

$HClO_3$: $+1 + x - 6 = 0,\ x = +5$

Final Answer : (2)

79)

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Explaining : Element is in its lowest oxidation state.
Concept : This question is based on oxidising and reducing agent.
Solution : Reducing agent
Final Answer : (1)

80)

Explaining : There is transfer of five electrons.
Concept : This question is based on balancing of redox reaction
Solution : $\overset{+7}{Mn}O_4^- \rightarrow \overset{+2}{Mn}$
Final Answer : (4)

81)

Explaining : In the following reaction.
Concept : This question is based on types of redox reaction.
Solution : It is disproportionation in which phosphorus is oxidised as well as reduced.
Final Answer : (2)

82)

Explaining : Oxidation state of Cr
Concept : This question is based on oxidation number
Solution :
$[Cr(NH_3)_4Cl_2]^+$
$x + 4(0) - 2 = +1$
$\Rightarrow x = +3$
Final Answer : (3)

83)

Explaining : Correct coefficients of the reactants
Concept : This question is based on balancing of redox reaction
Solution : -
$16H^+ + 2MnO_4^- + 10e^- \rightarrow 2Mn^{+2} + 8H_2O$
$5C_2O_4^{2-} \rightarrow 10CO_2 + 10e^-$

$16H^+ + 2MnO_4^- + 5C_2O_4^{2-} \rightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$
Final Answer : (2)

84)

Explaining : Value of n
Concept : This question is based on balancing of redox reaction
Solution :
$Cr_2O_7^{2-} + 14H^+ + nFe^{2+} \rightarrow 2Cr^{3+} + nFe^{3+} + 7H_2O$

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–2+14 ⇒ (+12)+6×2 = 24 (+6)+6×3 ⇒ 24
Final Answer : (2)

85)

Explaining : Increasing order of oxidation number
Concept : This question is based on oxidation number
Solution :
$NH_4^+ < N_2O < NO < NO_2 < NO_3^-$
Oxidation states of nitrogen respectively are –3, +1, +2, + 4, + 5
Final Answer : (2)

86)

Explaining : Stoichiometric cofficient
Concept : This question is based on balancing of redox reaction
Solution : The balanced chemical equation is :

$$3I_2 + 6OH^- \longrightarrow IO_3^- + 5I^- + 3H_2O$$

p = 3, q = 6, r = 5
Final Answer : (4)

87) **Question Explanation :** Effect of adding inert gas (He) at constant P and T on the dissociation equilibrium of PCl_5 and whether the given reason correctly explains the assertion.
Concept : This question is based on Le-Chatelier's principle and equilibrium constant dependence
Solution :
At constant pressure, adding He raises total volume → partial pressures of gaseous species fall → equilibrium shifts toward side with more gas moles (forward) so PCl_5 dissociation increases.
R is true: K changes only with temperature.
R does not explain A (A is due to change in partial pressures/volume, not change in K).
Final Answer: (1)

88)

Question Explanation : With the help of given list X and list Y, we need to find correct option.
Concept : Effect of temperature on equilibrium.
Solution :
(A) Active mass → Molar concentration →(II)
(B) Dynamic nature → Chemical equilibrium → (V)
(C) A + heat ⇌ B → forward reaction increases on increasing temperature → (IV)
(D) $\log\left(\frac{k_2}{k_1}\right) = \frac{\Delta H}{2.303}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$ ⇒ Vant Hoff's equation → (III)
(E) $2A_{(g)} + B_{(g)} \rightleftharpoons 3C_{(g)} \rightarrow \Delta n_g = 0 \rightarrow$ (I)
Correct match - A - (II), B - (V), C - (IV), D - (III), E - (I)
Final Answer : (3)

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NOVA

89)

Question Explanation: Assertion says $\Delta H_f°$ for $H_2(g)$ and $Cl_2(g)$ is zero; Reason says $\Delta H_f°$ for every element in its stable standard state is zero.
Concept: Standard enthalpy of formation
Solution:
By definition, standard enthalpy of formation of an element in its most stable form (standard state) at 1 atm and 298 K is taken as zero.
$H_2(g)$ and $Cl_2(g)$ are elements in their stable standard states, so their $\Delta H_f° = 0$. Hence both statements are true and (R) correctly explains (A).
Final Answer: (4)

90)

Question Explanation: The question asks for the truth value of two statements regarding the definition and examples of an isolated thermodynamic system.
Concept: Thermodynamic Systems Classification
Solution:
Analysis of Statement I: Statement I: In an isolated system there is no exchange of energy but exchange of matter is possible between system and the surroundings.

A. Definition of an isolated system: **No exchange of energy AND no exchange of matter.**

B. The statement incorrectly claims that the exchange of matter is possible.

C. Statement I is **Incorrect.**

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Analysis of Statement II: Statement II: The presence of reactants in a thermos flask is an example of an isolated system.

A. A thermos flask (or Dewar flask) is designed to minimize the exchange of both heat (energy) and matter with the surroundings, approximating an isolated system.

B. Statement II is **Correct.**

Final Answer : (3)

BIOLOGY

91)

Question Explanation : Asking which enzyme converts fibrinogen into fibrin during blood coagulation.

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Concept : Hematology – Blood coagulation
Solution : The correct answer is option 3 – Thrombin.

A. During **blood coagulation**, **fibrinogen** (inactive) is converted into **fibrin** (active) by the enzyme **thrombin**, forming a mesh that helps in clot formation.

Final Answer : Option 3

92) **Question is Asking About:**
Origin of heart beat and its conduction.

Solution/Explanation:
(SA node → AV node → Bundle of His → Purkinje fibre)
Final Answer : (4)

93) The pacemaker in a frog heart is the **Sinus venosus**.
Here's why:

A. The sinus venosus is a thin-walled chamber that receives blood from the veins.
B. It generates spontaneous electrical impulses that initiate the heartbeat.
C. These impulses spread through the heart, causing the atria and ventricles to contract in a coordinated manner.

94) **Problem Statement : Asking About :**
AB blood group can donate to.

Concept :
Blood Group.

Solution/Explanation :
A·B blood group is known to be universal receiver. As it has both A & B antigens it can only be donated to a AB blood group person who do not possess any antibody against these blood antigens.

95) **Problem Statement : Asking About :**
Correct about cardiac cycle.

Solution/Explanation :
During cardiac cycle volume of blood pumped out by each ventricle is same per beat & it is known as stroke volume.

96)
Question Explanation: Check if the reason correctly explains why high blood pressure causes damage to vital organs.

Concept: The question is based o**n** Blood pressure

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Solution: The correct answer is option 2.

A. **Assertion (A):** True. Hypertension damages cardiovascular and vital organ systems.
B. **Reason (R):** True. This is the diagnostic threshold.

Both Assertion and Reason are true, but the reason does NOT explain how hypertension leads to damage of heart, brain, and kidneys.
• Assertion talks about organ damage due to prolonged high BP.
• Reason only defines what value of BP is considered hypertension, not how it causes organ damage.
So, R is true but not the correct explanation of A.

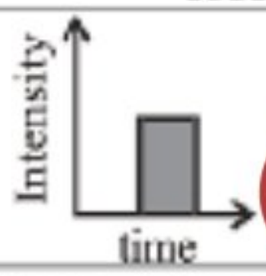
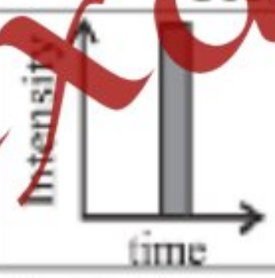
Final Answer: Option 2

97) **Problem Statement : Asking About :**
Mixing of oxygenated and deoxygenated blood.

Solution/Explanation :
Mixing of oxygenated & deoxygenated blood in ventricle where incomplete double circulation take place. i.e. Amphibian & reptiles.

98) **Problem Statement : Asking About :**
Correct about given figure.

Solution/Explanation :

	LUBB		DUB
A		A	
B	Due to closure of AV valves	B	Due to closure of SL valves
C	Systolic sound	C	Diastolic sound

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99) The correct answer is **Option 3: Three.**
Explanation:
In the human body, there are three organs that receive both oxygenated and deoxygenated blood:
1. Heart: The heart receives oxygenated blood from the lungs through the coronary arteries to supply its own muscle tissue with oxygen. It also receives deoxygenated blood from the veins that drain the heart muscle.
2. Liver: The liver receives oxygenated blood from the hepatic artery, which supplies it with oxygen and nutrients. It also receives deoxygenated blood from the portal vein, which carries blood from the digestive organs, containing nutrients absorbed from food.
3. Lungs: The lungs receive oxygenated blood from the bronchial arteries, which supply the lung tissue itself with oxygen. They also receive deoxygenated blood from the pulmonary artery, which carries blood from the heart to the lungs for oxygenation.
Therefore, the heart, liver, and lungs are the three organs in the human body that receive both oxygenated and deoxygenated blood.

100) **Question is Asking About:**

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Cardiac output is increased by

Solution/Explanation:
All factor which increases heart rate
(1) Sympathetic nerve Stimulation
(2) Adrenal medullary hormone
(3) Physical exercise

101) Thromboplastin activates prothrombin (in presence of Ca^{2+} and clotting factors), converting it into thrombin. Thrombin then converts soluble fibrinogen into insoluble fibrin threads forming the clot.

102) **Solution :** High blood pressure can affect vital organs like brain and kidney.
Correct Answer: (1)

103) **Solution :** The primary function of globulin proteins in blood plasma is to contribute to the body's defence mechanism by producing antibodies and participating in immune responses.
Final Answer : (3)

104) **Question is Asking About:**
Relationship between cardiac output heart rate and stroke volume is

Solution/Explanation:
Cardiac output = Stroke volume × Heart rate
Final Answer : (2)

105)

Question explanation
The question is asking which protein is found in the highest concentration in human blood plasma.

Concept
blood plasma.

Solution
Blood plasma contains three major types of proteins: albumin, globulin, and fibrinogen. Among these, albumin is the most abundant protein.

Final answer: (2)

106) **Solution :** In erythroblastosis foetalis, mother is Rh^- and father is Rh^+.
Final answer: (3)

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107) **Question Explanation:** Medical term for the sudden cessation of heartbeat.

Concept: Cardiac Arrest

Solution: Cardiac arrest is the condition where the heart completely stops beating, leading to a sudden loss of heart function, breathing, and consciousness.

A. **Angina** is chest pain due to inadequate oxygen.
B. **Heart failure** is the heart's inability to pump effectively.
C. **Heart block** is an electrical conduction problem, not necessarily a complete stop.

Final Answer: (2)

108) **Solution :** A → Prothrombin, B → Thrombin, C → Fibrinogen, D → Fibrin
Final answer: (2)

109)

Explain Question : Organism with single circulation of blood.
Concept : Types Of Blood Circulation And Types Of Heart.
Solution : In **fishes**, blood flows from heart → gills → body → back to heart — completing a **single circuit**. i.e. single circulation.
Final Answer : (1)

110) **Solution :** Lymph has WBCs, but no RBCs and platelets, Lymph lacks clotting capacity but contain fibrinogen.
Final Answer : (3)

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111) **Solution :** The portion of air which remains in the respiratory tract is called anatomical dead space. Its value for man is 150 ml.
Final Answer : (4)

112) **Solution :** pO_2 in deoxygenated blood = 40 mmHg
pCO_2 in oxygenated blood = 40 mmHg
Final Answer : (1)

113) **Solution :** Aquatic Arthopods, Molluscs, Fishes and tadpole respire through gills.
Final Answer : (2)

114) **Solution :** ERV is the amount of air expired over tidal volume by most forceful expiration.
Final Answer : (2)

115) **Solution :** A disorder in which serious lung damage occur due to proliferation of fibrous tissue is occupational respiratory disorder
Final Answer : (3)

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116) **Explain the Question :** Effect of CO_2 concentration on dissociation of oxyhaemoglobin is called ?
Concept : This question is based on Bohr's effect.
Solution : Effect of CO_2 concentration on dissociation of oxyhaemoglobin is called Bohr's effect.
Final Answer : (1)

117)

Question Explanation: Ion exchange in RBCs during CO_2 transport.
Concept: Transport of CO_2
Solution:

A. HCO_3^- (Bicarbonate ion) – Moves out of the red blood cell into plasma.
B. Cl^- (Chloride ion) – Moves into the red blood cell to maintain electrical neutrality.
C. This exchange is called the Chloride Shift (Hamburger phenomenon).

Final answer: (3)

118) **Explain Question:** Matching respiratory diseases with their features.
Concept: This question is based on respiratory disorders
Solution:
Asthma → allergy of bronchi/bronchioles.
Emphysema → damage of alveolar wall .
Occupational lung disease → fibrosis in lungs (e.g., silicosis, asbestosis).
CO_2 → Carbaminohemoglobin.
Bronchitis → inflammation of bronchi.
Correct matching:
A → iii, B → v ,C → i ,D → iv, E → ii
Final answer: (2)

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119)

Question Explanation: The question asks to identify the incorrect statement related to the oxyhaemoglobin dissociation curve and haemoglobin affinity.

Concept: The question is based on oxyhaemoglobin dissociation curve

Solution: The correct answer is option 4.

A. High pH and low CO_2 concentration cause left shift of oxyhaemoglobin dissociation curve → Correct.

B. Foetal haemoglobin (HbF) has more affinity for O_2 than adult haemoglobin (HbA) → Correct.

C. Oxyhaemoglobin dissociation curve is sigmoid in shape → Correct.

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Since all given statements are correct, there is no incorrect statement.

Final Answer: Option 4

120)

Question Explanation
The question is asking to arrange different respiratory volumes and capacities in increasing order based on their normal values.

Concept
Respiratory volumes and capacities

Solution

A. Tidal volume is the smallest, about 500 ml.
B. Residual volume is larger, about 1200 ml.
C. Inspiratory reserve volume is about 3000 ml.
D. Vital capacity is the largest, about 4500 ml.

So the increasing order is : A < B < C < D.

Final answer: Option 1

121)

Question Explanation: The question asks us to identify the variables represented on the axes of the oxygen–hemoglobin dissociation curve.

Concept: Oxygen dissociation curve.

Solution:

A. The curve is **sigmoidal** due to cooperative binding of oxygen to hemoglobin.

B. On the **x-axis**: partial pressure of oxygen (pO_2, mm Hg).

C. On the **y-axis**: percentage saturation of haemoglobin with oxygen.

Final Answer: (2)

122) **Question Explanation:** Factors affecting oxyhaemoglobin dissociation.
Concept: Oxygen Dissociation Curve
Solution: If blood pH falls (more acidic), hemoglobin's affinity for O_2 decreases → more O_2 is released.
Other options: pH rise, lower temperature, or lower BPG would decrease dissociation.

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Final Answer: (1)

123)

Question Explanation: Identify the correct statement about parts (A–D) in the circulatory diagram.

Concept: Exchange Of Gases

Solution: Systemic vein (deoxygenated): PO_2 = 40 mmHg, PCO_2 = 45 mmHg
Pulmonary artery: carries deoxygenated blood to lungs
Pulmonary vein: carries oxygenated blood from lungs to heart
Systemic artery: carries oxygenated blood to body tissues
Option (1) is correct because A (systemic vein) indeed has $PO_2 \approx 40$ mmHg and $PCO_2 \approx 45$ mmHg (values for deoxygenated blood returning from tissues).

Final answer: Option 1 – A is systemic vein with PO_2 = 40 mmHg, PCO_2 = 45 mmHg

124)

Explain Question : Put respiration events in the correct physiological order.
Concept : This question is based on Human Respiratory System (Nose And Nasal Chamber, Larynx)
Explanation :
(iii) Pulmonary ventilation: air enters lungs, CO_2 leaves.
(vi) Alveolar diffusion: $O_2 \leftrightarrow CO_2$ between alveoli & blood.
(ii) Transport by blood: gases carried to/from tissues.
(v) Tissue diffusion: gases exchange between blood & cells.
(i) Cellular utilisation of O_2: catabolism occurs, generating CO_2.
(iv) Release of resultant CO_2: CO_2 produced is released (ultimately exhaled).
Final answer : 1

125) **Question Explanation:** Chemoreceptors near the respiratory center and their sensitivity.
Concept: Regulation Of Respiration
Solution: The chemosensitive area responds strongly to CO_2 (via pH/H^+ changes), not O_2. Therefore, the correct combination is CO_2 and H^+.
Final Answer: Option (3)

126)

Explaining the Question :- Stimulus for breathing centre regulation

Concept:- This question is based on Regulation of respiration

Solution :- Option 3 is correct because an increase in CO_2 in arterial blood is the primary signal that stimulates the breathing center to initiate ventilation.

Final Answer :- 3 increase of CO_2 in arterial blood

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127)

Explain Question : Which statements about O_2 transport are true?
Concept : This question is based on Transport Of Oxygen
Explanation :
(A) Hb has 4 heme groups → max 4 O_2 molecules .Correct
(B) O_2 binding depends mainly on pO_2 .Correct
(C) 100 ml blood carries ~20 ml O_2 total, but delivers only ~5 ml to tissues (not 20 ml) .Incorrect
(D) O_2 dissociation curve is sigmoid, not hyperbola .Incorrect
Final answer : 3

128)

Explain Question:
The question asks which factors favour oxyhaemoglobin formation in the lungs (increase Hb's affinity for O_2)

Concept:
Oxyhaemoglobin formation and factors affecting Hb–O_2 binding

Solution:
Oxyhaemoglobin formation is favoured by:
• High O_2 partial pressure$PO_2 \uparrow$)
• Low CO_2 partial pressure ($PCO_2 \downarrow$)
• Low temperature (Temperature $\downarrow$)
• Low H^+ concentration (high pH, $H^+ \downarrow$)

Final answer: Option (4)

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129) **Explain Question:**
This question is asking about the correctness of statements related to parts of the human respiratory system and to determine how many statements are correct.

Concept:
Breathing and Exchange of Gases

Solution:
• Statements A is correct: the larynx is a cartilaginous box responsible for sound production.
• Statement B is incorrect: the epiglottis is made of elastic cartilage, not white fibrous cartilage.
• Statement C is incorrect: the trachea divides into bronchi at the level of the 5th thoracic vertebra, not the 6th.
• Statement D is correct: the pharynx is a common passage for food and air.

Final Answer: Option (3)

130) **Question Explanation:** Correctly identified lung part with its function.

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Concept: Exchange Of Gases

Solution: The alveolar cavity (A) is the main site where O_2 diffuses into blood and CO_2 diffuses out. This matches with option (3): "A : alveolar cavity – main site of exchange of respiratory gases."

Final answer: Option 3 – A : alveolar cavity, main site of exchange of respiratory gases

131) **Solution :** (I) The structure replicates during mitosis and generates the spindle–L
(II) Major site for synthesis of lipid–B
(III) Power house of the cell –H
(IV) Store house of digestive enzyme – J.
(V) Increase the surface area for the absorption of materials –N
(VI) Site of glycolysis – F
(VII) Site for active ribosomal RNA synthesis-D
Final Answer : (3)

132) **Solution : Statement III is incorrect.** A well-developed **cytoskeleton**, composed of microtubules, microfilaments, and intermediate filaments, is a characteristic feature of **eukaryotic cells**, providing structural support and facilitating movement.
Final Answer : (2)

133) A cyanobacterial cell. The question describes the entity as a full cell with core prokaryotic features and an internal photosynthetic membrane system, which is characteristic of cyanobacteria. Chloroplasts, though similar due to their cyanobacterial ancestry, exist within plant cells and do not function as free-living cells in modern plants.

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134) Presence of selectively permeable nuclear pores for macromolecule exchange is the unique characteristic. Nuclear pores are specialized gateways that allow regulated passage of large complexes like ribosomal subunits, various RNAs, and nuclear proteins. This ensures that transcription and DNA replication, which occur inside the nucleus, are tightly coordinated with translation and other cytoplasmic processes.

135) **Solution :** RBCs are specialized for respiratory gas transport, while sieve tube elements are specialized for translocation of organic solutes in plants. Bacterial cell contains a nucleoid while parenchyma cells contains chlorophyll for photosynthesis.
Final Answer : (2)

136) **Solution : Assertion (A)** is correct. **Lysosomes** are membrane-bound organelles that contain a variety of digestive enzymes. If the cell is damaged or breaks down, the lysosome's membrane can rupture, releasing these enzymes and causing the cell to digest itself, hence the name "suicidal bags."
Reason (R) is also correct. Lysosomes are rich in **hydrolytic enzymes**, such as lipases, proteases, and carbohydrases, which function optimally in an acidic environment (around pH 5) created by proton pumps in their membrane. The reason provides the specific biochemical basis for the assertion, explaining why lysosomes have this destructive capability.

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Final Answer : (1)

137) **Solution :** The physiochemical approach to study and understand living organisms is called reductionist biology.
Final Answer : (3)

138) **Solution :** Prokaryotes do not possess single inner chromosome
Final Answer : (1)

139) **Solution :** Labelling appears first in ribosomes bound to the RER, then in the Golgi, and finally in secretory vesicles, indicating a pathway where proteins are synthesized in the RER and then processed and packaged in the Golgi before secretion.
Final Answer : (4)

140) **Solution :** Only A and C are correct
Final Answer : (3)

141) **Solution :** Chloroplasts and mitochondria.
Final Answer : (4)

142) **Solution:** Bacterial genomic DNA is typically a single **circular** chromosome, not linear.
Final Answer : (3)

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143) **Solution:** Statement II is incorrect. Mitochondria, chloroplasts, and peroxisomes are not part of the endomembrane system.
Final Answer : (2)

144) **Solution:** The quasi-fluid lipid bilayer allows lateral movement of proteins.
Final Answer : (2)

145) **Solution:** Telocentric chromosomes have a **terminal centromere**, not an absent one.
Final Answer : (4)

146) **Solution :** In some bacteria, **fimbriae** are known to help attach the bacteria to rocks in streams and also to the host tissues.
Final Answer : (3)

147) **Solution :** They are hair-like outgrowths of the cell membrane.
Final Answer : (1)

148) **Solution : The Assertion is true:** The cell is indeed the fundamental structural and functional unit of all living organisms.

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The Reason is true: A cell contains all the necessary "machinery" (organelles, genetic material, and metabolic pathways) to perform the functions of life. Sub-cellular components (like a mitochondria or a nucleus on its own) cannot survive or reproduce independently.
The Connection: Because nothing simpler than a cell can live independently, the cell is defined as the "basic unit" of life.
Final Answer : (4)

149)

Solution : (A) Thylakoids — (III): These are flattened sacs found in the stroma of chloroplasts where the light-dependent reactions of photosynthesis occur.
(B) Cristae — (IV): These are the inner foldings of the mitochondrial membrane that increase surface area for ATP production.
(C) Cisternae — (I): These are the flattened, disc-like sacs that stack together to form the Golgi apparatus.
(D) Chromatin — (II): This is the complex of DNA and proteins (nucleoproteins) found in the nucleus that condenses to form chromosomes.
Final Answer : (3)

150) **Solution :** This feature correctly aligns with plant cells that have a cellulose cell wall, large central vacuole, and chloroplasts.
Presence of plasmodesmata connecting adjacent cells. Plant cells typically have rigid cell walls made of cellulose, large central vacuoles maintaining turgor pressure, and chloroplasts for photosynthesis. In addition, neighboring plant cells are connected by cytoplasmic strands called plasmodesmata that pass through the cell walls, allowing transport of ions, small molecules, and signaling substances between cells.
Final Answer : (2)

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151) **Solution :** Meiosis involve recombination which lead to genetic variability in the population of organisms from one generation to next.
Final Answer : (1)

152) **Solution :** In metaphase of mitosis, spindle fibres attaches to the kinetochore of chromosomes. Chromosomes are moved to spindle equator and get aligned along metaphasic plate through spindle fibres of both poles.
Final Answer : (3)

153) **Solution :** Unlike animal cells (which use a furrow/ centripetal method), plant cells form a **cell plate** starting in the **center** and growing **centrifugally** (outward). This cell plate eventually becomes the **middle lamella**, the cementing layer between the primary walls of two neighboring plant cells.
Final Answer : (2)

154) **Solution :** (iv) Chromosomes move to the spindle equator metaphase stage.
(v) Crossing over occurs in homologous chromosomes.
Final Answer : (3)

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155) **Solution :** A–IV, B–I, C–II, D–III

Column II		Column II	
A.	Centromere splits and chromatids separate apart	IV.	Anaphase
B.	Pairing between homologous chromosomes takes place	I.	Zygotene
C.	Chiasmata	II.	Point of attachment between homologous chromosome
D.	Synaptonemal complex	III.	Nuclear protein complex that helps in adherence of non-sister chromatids of homologous chromosomes.

Final Answer : (3)

156) **Solution :** The first meiotic division leads to the reduction of chromosome number to half and the second meiotic division segregates the replicated chromosomes (2C → C).
Final Answer : (4)

157) **Solution :** In meiotic division, metaphase-I spindle starts appearing and bivalents becomes attached to spindle through centromeres. Bivalents, then appear in the form of an equatorial plate due to the movement known as congression.
Final Answer : (1)

158) **Solution :** Crossing over is an enzyme-mediated process and the enzyme involved is called recombinase.
Final Answer : (3)

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159) **Solution :** (ii) During this time the amount of DNA per cell double.
(vi) G_0 phase is called as quiescent stage.
Final Answer : (2)

160) **Solution :** Sufficient cell size, accumulation of cyclin–CDK complexes, and favorable external signals. The G_1/S checkpoint is controlled by regulatory proteins (cyclins and CDKs) and checks for adequate cell growth, energy, and DNA integrity before DNA replication begins. Spindle formation occurs much later, after DNA has already been replicated in S phase and condensed in prophase, so it cannot define the transition from G_1 to S phase.
Final Answer : (3)

161) **Solution :**
I. True – Enzymes lower activation energy to speed up reactions.
II. False – Cofactors are required by many types of enzymes, not only oxidoreductases.
III. True – Enzyme specificity depends on the complementary 3D shape of the active site.
Final Answer: (2)

162) **Solution :** I. True – Enzymes remain chemically unchanged after the reaction.

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II. True – Association of coenzyme with apoenzyme is only transit and occur during thee course of catalysis.
III. False – Competitive inhibitors do not permanently destroy the enzyme; they temporarily block the active site.
Final Answer : (1)

163)

Solution: When substrate concentration is very low, fewer enzyme active sites are occupied. This reduces the rate of the reaction because the enzyme cannot work at full capacity.
The enzyme itself is not denatured, converted, or permanently affected.
Final Answer : (1)

164) **Solution :** H_2O > Proteins > Nucleic acid > Carbohydrates > Lipids > Ions.
Final Answer : (2)

165) **Solution :** Starch forms helical secondary structures.
Paper made from plant pulp and cotton fibre is cellulosic.
Final Answer : (2)

166) **Solution :** In solution of different pH, the structure of amino acids changes because a particular property of amino acids is the ionizable nature of $-NH_2$ and –COOH.
Final Answer : (1)

167) **Concept :** Metal ions as cofactors.
Solution: A number of enzymes require metal ions for their activity. Zinc is a cofactor for the proteolytic enzyme carboxypeptidase.
Final Answer: (2)

168) **Concept :** Enzyme classification (Class III).
Solution: Hydrolases (Class III) catalyze the hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds.
Final Answer: (3)

169) **Concept:** Types of enzyme inhibition.
Solution: Malonate closely resembles the substrate succinate in structure and competes for the active site of succinate dehydrogenase, which is the definition of competitive inhibition.
Final Answer: (3)

170) **Concept :** Classification of co-factors (Prosthetic groups).
Solution: identifies 'Haem' as a prosthetic group in enzymes like peroxidase and catalase. NAD and NADP are co-enzymes, and Zinc is a metal ion cofactor.
Final Answer: (3)

171) **Solution :** (a) Tryptophan → (v) Aromatic amino acid

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(b) Lysine → (iv) Basic amino acid
(c) Valine → (ii) Neutral amino acid
(d) Serine → (iii) Hydroxy methyl as R group
Final Answer : (4)

172) **Solution :** (a) In proteins, only right handed helices are observed.
(c) In primary structure of protein, the last amino acid is called C-terminal amino acid.
(d) Two subunits of α-type and two subunits of β-type together constitute the human haemoglobin (Hb).
Final Answer : (2)

173) **Solution :** Both A and R are true but R is not a correct explanation of A.
Final Answer : (2)

174) **Solution :** A = Adenine (N-base);
B = Adenosine (Nucleoside);
C = Adenylic acid (Nucleotide)
Final Answer : (2)

175) **Solution :** G.N. Ramachandran discovered the triple helical structure of collagen.
Final Answer : (4)

176) **Solution :** (a) Glycerol → (iii) Simple lipid
(b) Palmitic acid → (i) Saturated fatty acid
(c) Lecithin → (ii) Phospholipid
(d) Arachidonic acid → (iv) Unsaturated fatty acid
Final Answer : (2)

177) **Concept:** Non-reducing sugar
Solution: Sucrose has no free aldehyde/ketone group
Final Answer: (1)

178) **Concept:** Chitin structure.
Solution: • Correct Option (3): Describes complex polysaccharides like chitin as often being homopolymers (specifically of substituted sugars like N-acetyl glucosamine).
• **Incorrect Options:** (1) describes cellulose/starch. (2) Chitin is considered a homopolymer of the modified sugar unit. (4) Chitin is a structural, not storage, polysaccharide.
Final Answer: (3)

179) **Solution :** a–iv, b–v, c–iii, d–ii, e–i

(a)	Ligases	(vi)	Joining of C–O, C–S, C–N, P–O bonds
(b)	Transferases	(v)	S–G + S' → S + S'–G
(c)	Lyases	(iii)	Leaving double bonds

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(d)	Oxid-reductases	(ii)	S-reduced + S'-oxidised → S-oxidised + S'-reduced
(e)	Hydrolases	(i)	Hydrolysis of C–C, C-halide or P–N bonds

Final Answer : (3)

180) **Concept:** Examples of Fatty Acids.
Solution: Specifies Palmitic acid has 16 carbons including the carboxyl carbon, while Arachidonic acid has 20 carbons including the carboxyl carbon.
Final Answer: (2)

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